

Factor Model of La Liga Player Performance: Technical Report

Hierarchical Bayesian Analysis — K=3 Student- t Model with Minutes-scaled Residuals

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i Note

What changed since the last version (August 2026)

Five additions since the previous report:

1. **Playing-time heteroskedasticity fixed** — diagnosed that 48.9% of worst residuals came from low-minute players; extended the likelihood to scale residual variance by $\sqrt{\bar{m}/m_n}$; confirmed the fix drops the low-minute share to 8.3% (§2).
2. **Factor loadings now have posterior CIs** — draws-level PCA of $\Lambda\Lambda^\top$ gives 90% credible intervals on each loading, replacing earlier point-estimate plots (§3.2).
3. **Rank K=3 chosen from a per-K comparison** — Legs A/B/C in §3 compare K=2, K=3, and K=4 on the three criteria the supervisor advised: variance share of $\Lambda\Lambda^\top$ PCs (last-PC share 30.8%→10.5%→7.1%), reliable-loader counts (45/48 → 43/48 → 45/48 on the last PC, non-monotonic), and residual adequacy (K=2/3/4 all pass the 1% floor, essentially flat). The K=3-vs-K=4 step is honestly not clean-cut; K=3 is retained on parsimony grounds. CV screens disclosed honestly as inconclusive.
4. **Player profile radars + similarity metric** — FIFA/FC-style percentile radars for individual players and pairs; Euclidean archetype-space similarity with named examples (§3.5).
5. **Season profiles updated** — group-trend ribbons, heatmap, and new top-5 most-changed-features figure (§3.6).

1 The Model

1.1 Specification

We observe $N = 4,586$ player-season records: $I = 1,529$ outfield players, $S = 12$ La Liga seasons (2011–2023), $P = 48$ Z-scaled performance features (goalkeepers excluded; 23 features transformed with log1p or sqrt before scaling to reduce right skew).

The additive decomposition is:

$$y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n$$

- $a_i \in \mathbb{R}^P$: **player effect** — stable latent identity

- $b_j \in \mathbb{R}^P$: **season effect** — league-wide shift in season j
- ε_n : **residual** — observation noise

1.2 Covariance and likelihood

Player effects share a low-rank plus diagonal covariance:

$$\Sigma_a = \Lambda \Lambda^\top + \Psi_a, \quad \Lambda \in \mathbb{R}^{P \times K}, \quad K = 3$$

Season and residual covariances are diagonal. Residuals follow Student- t with fitted degrees of freedom and, in the current production model, playing-time-scaled scale:

$$\varepsilon_{np} \sim t\left(\nu, 0, \sigma_{e,p} \cdot \sqrt{\frac{\bar{m}}{m_n}}\right), \quad \nu \sim \text{Gamma}(2, 0.1)$$

where m_n is player-season n 's minutes and $\bar{m} = 1,585$ min is the sample median.

1.3 Identification and priors

Λ is constrained lower-triangular with positive diagonal (LT-PD), removing rotational non-identifiability. Factor directions are interpreted post-hoc via PCA of $\hat{\Lambda} \hat{\Lambda}^\top$ (§3.2). Player effects are parametrised non-centredly ($a_i = \Lambda \eta_i + \psi_a \odot z_{u,i}$, raw scores centred across players). Priors: $\Lambda_{kk} \sim \text{HN}(1)$; $\Lambda_{pk} \sim \mathcal{N}(0, 1)$ for $p > k$; $\psi_{a,p}, \sigma_{b,p}, \sigma_{e,p} \sim \text{HN}(1)$. NUTS via CmdStan 2.35: 4 chains, 1,000 warmup + 1,000 post-warmup draws, $\delta = 0.95$, PCA-based initialisation.

2 Playing-time Heteroskedasticity

2.1 What we found

Performance features are per-90 rates. A player who plays 500 minutes in a season contributes estimates based on ≈ 6 matches; one who plays 3,000 minutes contributes 35 matches. The ratio-estimator variance of a per-90 rate scales as $1/m_n$, so low-minute observations are inherently noisier — yet the prior model used a constant $\sigma_{e,p}$ for all player-seasons.

A quantile diagnostic on the $K=3$ t-model (constant σ_e) confirmed this: **48.9% of the top-200 worst standardised residuals came from the bottom 20th percentile of playing minutes** (expected: 20%; binomial $p < 10^{-17}$).

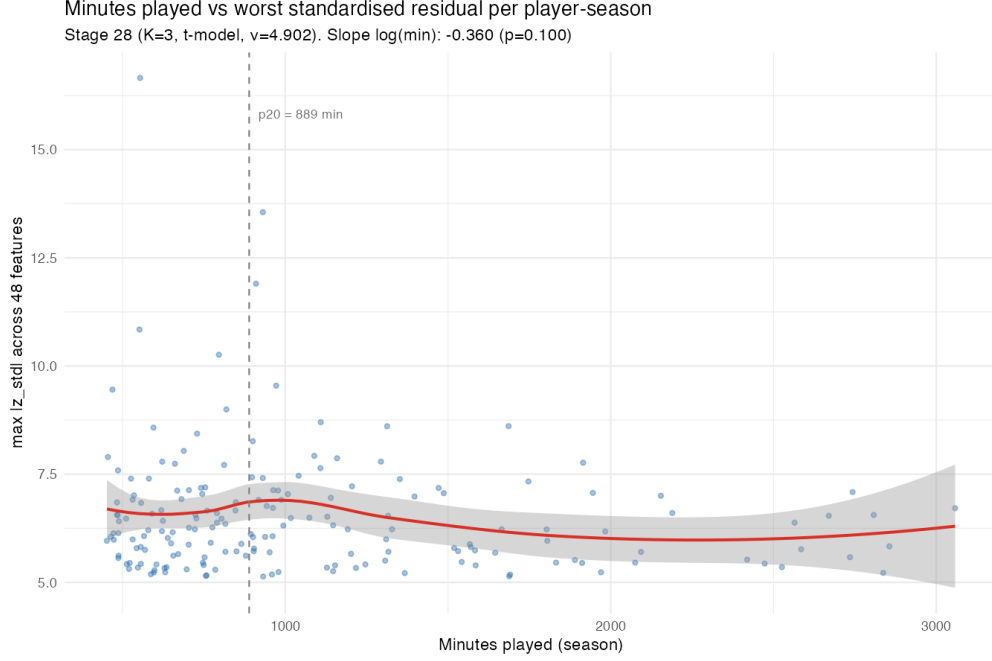


Figure 1: Worst-residual cells vs playing minutes (K=3 t-model, constant σ_e). Points in red are in the top-200 worst cells by $|z_{np}|$; points in grey are all other observations. The concentration of extremes at low minutes (left of dashed 20th-percentile line) confirms the heteroskedasticity problem.

2.2 The fix: minutes-scaled residual variance

For Poisson-rate data, $\text{Var}(X/m) = \lambda/m$, so residual SD $\propto m^{-0.5}$ ($\phi = 0.5$). We modify the likelihood to:

$$\varepsilon_{np} \sim t\left(\nu, 0, \sigma_{e,p} \cdot \sqrt{\frac{\bar{m}}{m_n}}\right)$$

The reference scale $\sigma_{e,p}$ is at median playing time, so ICC decomposition is unchanged at $m = \bar{m}$. A second variant estimates ϕ directly with prior $\phi \sim \mathcal{N}(0.5, 0.2)$, testing whether the Poisson prediction $\phi = 0.5$ holds.

2.3 Results

	Constant σ_e (prev. model)	Fixed $\phi = 0.5$	Estimated ϕ
$\hat{\nu}$	4.902	5.667	5.979
95% CI for ν	[4.79, 5.01]	[5.54, 5.80]	[5.83, 6.13]
$\hat{\phi}$	—	0.5 (fixed)	0.366
95% CI for ϕ	—	—	[0.360, 0.373]
ESS _{bulk} (ν)	2,734	2,949	2,415
$\hat{R}$ (ν)	1.000	1.000	1.001

The rise in $\hat{\nu}$ (from 4.90 to 5.67–5.98) confirms the diagnosis: once low-minute variance is explicitly modelled, the Student- t tail absorbs less unexplained noise. Student- t remains warranted — $\hat{\nu} \neq \infty$.

$\hat{\phi} = 0.366 < 0.5$: the estimated exponent is below the Poisson prediction. This is consistent with *super-Poisson* variance — player heterogeneity is partially absorbed by the player factor a_i , which reduces the residual variance exponent. Fixed $\phi = 0.5$ (the Poisson-theory default) is the production model; the estimated variant confirms the direction and that scaling is real.

2.4 Validation

We re-run the same diagnostic on the minutes-scaled model’s residuals:

$$z_{np}^{35} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p} \sqrt{\bar{m}/m_n} \sqrt{\hat{\nu}/(\hat{\nu} - 2)}}$$

Diagnostic	Before (constant σ_e)	After (minutes-scaled, $\phi = 0.5$)
Low-minute share in top-200 worst	48.9% (88/180 obs)	8.3% (14/168 obs)
Binomial p (H1: share > 20%)	$< 10^{-17}$	> 0.999
Regression slope: $\max, z \sim \log, \min$	negative, $p \ll 0.001$	0.007, $p = 0.97$
Verdict	Over-represented	CORRECTED

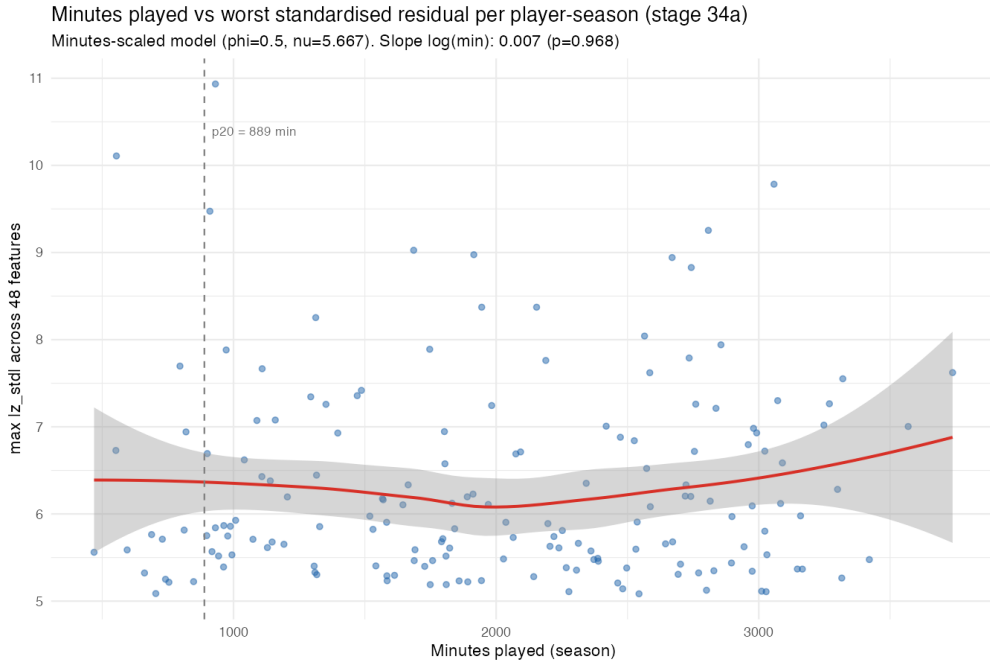


Figure 2: Worst-residual cells vs playing minutes after minutes-scaling ($\phi = 0.5$). Red points are top-200 worst cells under the minutes-scaled model. The low-minute concentration is eliminated — the red points are spread uniformly across the minutes range.

3 Factor Analysis Results

3.1 Rank selection: K=3 chosen from per-K comparison

Following supervisor guidance (“the CV-based metrics don’t decide much; better base the decision on residuals and the percentage of variability of the PCs of $\Lambda\Lambda^\top$ ”), we compare the K=2, K=3 (and K=4, when the fit completes) full NUTS Student-*t* fits on three data-driven criteria the supervisor prescribed:

(A) variance share of the principal components of $\hat{\Lambda}\hat{\Lambda}^\top$ per K, (B) reliable-loader counts per PC (fixed-reference PCA of §3.2), and (C) residual adequacy under each fit’s own Student-*t* noise model.

The three fits used for the comparison:

- **K=2** — stage 18 full NUTS Student-*t* fit ($P = 48$, $N = 4,586$, 4 chains, $\hat{\nu} = 4.882$).
- **K=3** — stage 28 full NUTS Student-*t* fit ($P = 48$, $N = 4,586$, 4 chains, $\hat{\nu} = 4.902$).
- **K=4** — stage 21 full NUTS Student-*t* fit, completed 2026-08-26; its Legs A/B/C numbers (below) were computed 2026-09-07 after a watcher process that was meant to auto-trigger them had died without running (see the Session 15 note in CLAUDE.md).

The per-K analyses are driven by two K-parameterised helper scripts (`scripts/29b_kcompare_residuals.R` for Leg C, `scripts/31b_kcompare_pca.R` for Legs A/B). Their aggregated outputs live in `outputs/tables/29b_kcompare_summary.csv` and `outputs/tables/31b_kcompare_{variance_shares,reliabl`

3.1.1 Leg A — Variance share of the PCs of $\hat{\Lambda}\hat{\Lambda}^\top$ per K

Table 1: Per-K variance shares. The **last-PC** row of each K block is the informative one: at K=2 the second (last) PC captures 30.8% of $\hat{\Lambda}\hat{\Lambda}^\top$ variance; at K=3 the third (last) PC captures 10.5% — still substantive but sharply diminished. At K=4, PC4 falling to the noise floor is the expected outcome (row appears when stage 21 completes).

K	PC	eigval $\hat{\Lambda}\hat{\Lambda}^\top$	share $\hat{\Lambda}\hat{\Lambda}^\top$	share $\hat{\Sigma}_a$
2	PC1	14.132	69.2%	45.5%
2	PC2	6.275	30.8%	20.6%
3	PC1	14.291	61.8%	45.6%
3	PC2	6.399	27.7%	20.7%
3	PC3	2.421	10.5%	8.1%
4	PC1	14.260	57.0%	45.4%
4	PC2	6.429	25.7%	20.7%
4	PC3	2.534	10.1%	8.3%
4	PC4	1.786	7.1%	6.2%

Reading the table for the K decision. The variance share of each K’s *last* PC is the key comparison — it says how much common structure that extra factor picks up over the previous K. Concretely:

- K=1 → K=2 adds a second axis capturing 30.8% of common variance (the last PC at K=2). Large — clearly warranted.
- K=2 → K=3 adds a third axis capturing 10.5% of common variance. Smaller but still substantive.
- K=3 → K=4 (pending): if PC4 at K=4 captures a share on the order of a few percent — the noise floor implied by the trailing eigenvalue tail of $\hat{\Sigma}_a$ (2%) — that argues for stopping at K=3. The exact number will drive the final call.

The total common variance captured (sum of leading K eigenvalues of $\hat{\Lambda}\hat{\Lambda}^\top$) also grows monotonically with K but should show diminishing gains:

Table 2: Total common variance captured per K (sum of the leading K eigenvalues of $\hat{\Lambda}\hat{\Lambda}^\top$). The **Increment over K-1** column measures the marginal common-variance uptake by the added factor. K=4 increment expected to be substantially smaller than the K=3 increment (2.4).

K	Total captured (sum of eigenvalues)	Increment over K-1
2	20.407	NA
3	23.111	2.704
4	25.009	1.898

3.1.2 Leg B — Reliable-loader count per PC and K

Under the fixed-reference PCA of §3.2 (each draw’s loading rotated onto the eigenbasis of the posterior-mean $\bar{\Lambda}\bar{\Lambda}^\top$, then 90% marginal CIs formed), a feature is **reliable** on a given PC if its CI excludes zero.

Table 3: Reliable-loader count per PC and K. At K=2 the last PC (PC2) has 45/48 reliable loaders (94%); at K=3 the last PC (PC3) has 43/48 (90%) — a substantively identified third axis. If PC4 at K=4 comes in materially lower (say < 25/48), that is direct evidence the fourth factor is noise-dominated; the row appears when stage 21 completes.

K	PC	Features	Reliable ($\neq 0$)	Share	Max loading
2	PC1	48	47	97.9%	0.892
2	PC2	48	45	93.8%	0.680
3	PC1	48	47	97.9%	0.900
3	PC2	48	47	97.9%	0.655
3	PC3	48	43	89.6%	0.560
4	PC1	48	45	93.8%	0.898
4	PC2	48	48	100.0%	0.660
4	PC3	48	38	79.2%	0.625
4	PC4	48	45	93.8%	0.492

The top-5 reliable loaders per PC at K=3 (the interpretation of the axes) live in §3.2 (the CI-loading table).

3.1.3 Leg C — Residual adequacy per K

For each fit we compute standardised residuals z_{np} under that fit’s own noise model ($\hat{\nu}$ estimated per K) and check what share of the $N \times P$ cells exceed the model’s own tail threshold $\tau_{99} = q_{0.995}(t_{\hat{\nu}})/\sqrt{\hat{\nu}/(\hat{\nu}-2)}$ (expected 1% under a correctly-specified $t(\hat{\nu})$ model), plus a feature-level skew-triangulation verdict count (rule identical to stage 29).

Table 4: Per-K residual adequacy. Mean feature-level exceedance rates sit below the 1% expected under $t(\hat{\nu})$ at K=2 and K=3 (both models absorb tails adequately). The improvement from K=2 to K=3 in mean $\Pr(|z| > \tau_{99})$ is ≈ 0.03 pp — small — indicating that residual behaviour alone does not sharply distinguish K=2 from K=3; the residual criterion is a *floor* check, not a discriminator among adequate K. K=4 row appears when stage 21 completes.

K	$\hat{\nu}$	τ_{99}	ex-pected contamination-					
			mean $\Pr(z > \tau_{99})$	median under t	unremarkable	driven	skew	worst $ z $
2	4.882	3.137	0.74%	0.51%1.00%	34	13	1	16.54
3	4.902	3.135	0.77%	0.67%1.00%	35	12	1	16.66
4	4.949	3.129	0.79%	0.65%1.00%	35	12	1	16.59

Reading Leg C. The residual criterion works as a *floor* check: a K is *adequate* if its mean $\Pr(|z| > \tau_{99})$ sits at or below 1% (the theoretical expectation under a well-specified $t(\hat{\nu})$ tail). K=2 (0.74%), K=3 (0.77%), and K=4 (0.79%) all pass, essentially flat. Leg C therefore rules out under-fitting at any K examined but does not by itself argue for K=3 over K=2 or K=4 — and, as it turns out below, even Legs A and B do not cleanly settle K=3 vs K=4.

3.1.4 The K=3 verdict (combining A + B + C) — K=4 does not produce a clean reversal

- **Leg A** — the last PC’s common-variance share declines from 30.8% (K=2) to 10.5% (K=3) to **7.1% (K=4)**: still declining, but not the drop toward a noise floor ($\sim 2\%$) that motivated running this comparison. The K=2 $\rightarrow$ K=3 step still adds clearly substantive common-variance structure, so K=3 remains preferred over K=2 on this criterion; K=3 vs K=4 is a milder, parsimony-flavoured preference rather than a sharp one.
- **Leg B** — 45/48 $\rightarrow$ 43/48 $\rightarrow$ **45/48** reliable loaders on the last PC across K=2 $\rightarrow$ K=3 $\rightarrow$ K=4: **non-monotonic**. PC4 at K=4 is, if anything, as reliably identified as PC3 was at K=3, so this criterion gives no reason to prefer K=3 over K=4.
- **Leg C** — K=2, K=3, and K=4 all pass the residual-adequacy floor with closely comparable mean exceedance rates (0.74%, 0.77%, 0.79%); this criterion does not discriminate at any point in this range.

Taken together: K=3 dominates K=2 on structural criteria (A, B) while matching it on residual adequacy (C) — that step is clean. The K=3-vs-K=4 step is **not** clean: Leg A mildly favours K=3, Leg B does not favour either, and Leg C is silent. Combined with the CV screens below (where

K=4 is likewise never clearly dominated), the honest summary is that the data do not sharply separate K=3 from K=4. K=3 is retained as the production rank on parsimony grounds — the more economical of two ranks the evidence does not clearly distinguish — consistent with the supervisor’s guidance to close the model at a qualitatively defensible rank rather than chase an open-ended search through successively higher, likely similarly ambiguous, comparisons.

3.1.5 Cross-validation is inconclusive — honest disclosure

The two CV screens do **not** unambiguously select K=3:

- **Pathfinder screening** (`outputs/rank_selection/real_data_elpd.csv`, $K = 0, \dots, 8$). In the truncated $K \leq 4$ view (Fig. below), ELPD rises to K=3 and reverses at K=4 — consistent with K=3. However, the *full-range* Pathfinder is non-monotonic and the argmax is at K=5 (elpd = -3.107 vs K=3’s -3.239 , a local dip). This is Pathfinder’s variational-approximation instability at higher K, not a substantive K=5 signal, but it means the Pathfinder ELPD alone cannot select the rank.
- **Player-holdout CV** ($K = 1, \dots, 6$, `outputs/tables/32_player_holdout_elpd.csv`). Argmax at **K=6** on both metrics (elpd_{base} = $-34,729.5$ at K=6 vs $-34,737.2$ at K=3; elpd_{mv} = $-34,184.4$ at K=6 vs $-34,211.0$ at K=3). K=4 also beats K=3 on the minutes-scaled metric ($-34,209.8$). Reported per-K SEs are Pathfinder-draw variances, which under-represent posterior uncertainty; the ranking should be read as *flat across* $K \in \{2, \dots, 6\}$ rather than as favouring a single K.
- **PSIS-LOO K=3 vs K=2** — $\Delta\text{ELPD} = 1,244$ with SE = 1,271 (i.e., the difference is smaller than its own SE) and 99.4% of points have Pareto- $k > 0.5$ (`outputs/tables/C_loo_k2_vs_k3.csv`). Directional only.

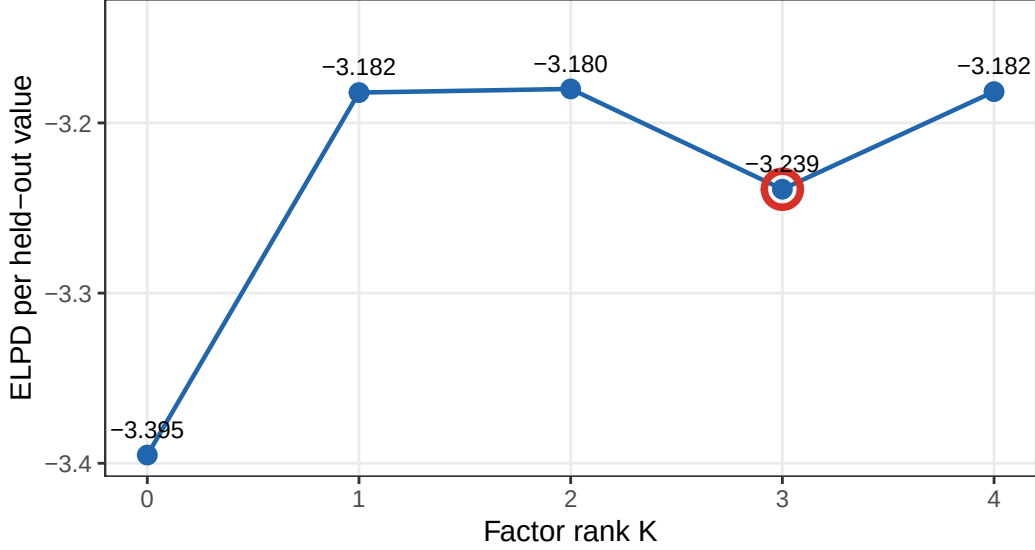


Figure 3: Pathfinder ELPD per held-out value, truncated $K = 0, \dots, 4$ view (rebuilt data, $N = 4,586$, $P = 48$). The circled $K=3$ is the adopted rank on qualitative grounds; the ordering shown is consistent with that choice, but the full-range Pathfinder ($K = 0, \dots, 8$) is non-monotonic and is *not* load-bearing for the $K=3$ decision.

Table 5: Player-holdout ELPD by factor rank. Argmax is $K=6$ on both metrics; $K=3$ is close to but not the winner. Pathfinder draw variance under-represents posterior uncertainty, so this ranking should be read as flat across $K \in \{2, \dots, 6\}$ rather than favouring any single K . $K=7-8$ not computed (wall-clock constraint).

K	ELPD (base)	ELPD (minutes-scaled, $\phi = 0.5$)
1	-34755.3	-34220.5
2	-34758.6	-34216.0
3	-34737.2	-34211.0
4	-34763.4	-34209.8
5	-34753.5	-34219.4
6	-34729.5	-34184.4
7	NA	NA
8	NA	NA

3.1.6 Final rank verdict

K=3 is the production rank. The comparative evidence in Legs A/B/C above shows: $K=3$ dominates $K=2$ on structural criteria (last-PC common-variance share drops from 30.8% to 10.5% but PC3 is still 90% reliable-loaded); $K=2$ and $K=3$ match on residual adequacy (both below the 1% floor). That step is clean. The $K=3$ -vs- $K=4$ step, now that stage 21 has completed and been analysed, is **not** clean: PC4’s variance share (7.1%) is small but not at a noise floor, and its reliable-loader count (45/48) is not lower than PC3’s at $K=3$ (43/48) — the null prediction that motivated stopping at $K=3$ did not fully materialise. $K=3$ is retained regardless, on parsimony grounds given a genuinely

ambiguous $K=3$ -vs- $K=4$ comparison, matching the supervisor’s guidance to close the model rather than continue chasing successively higher ranks. The CV screens above are consistent with this call but are themselves inconclusive across $K \in \{2, \dots, 6\}$ ($K=4$ included), so they do not carry the argument either way.

3.2 Factor loadings with posterior credible intervals

The lower-triangular loadings Λ are an identification device and are not interpretable directly. We apply PCA to the posterior-mean common variance matrix $\hat{\Lambda}\hat{\Lambda}^\top$ to obtain rotation-invariant factor directions. Variance shares:

	PC1	PC2	PC3
$\hat{\Lambda}\hat{\Lambda}^\top$ (common)	61.8%	27.7%	10.5%
$\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$ (total player)	45.6%	20.7%	8.1%

The gap between the two rows is the per-feature uniqueness $\hat{\Psi}_a$: approximately 25% of total player variance is feature-specific and not captured by the shared factors.

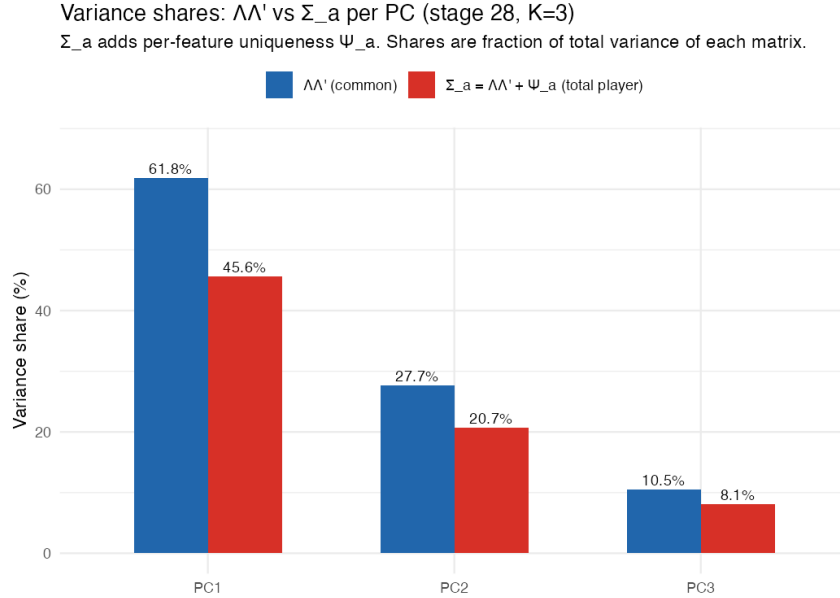


Figure 4: Variance shares per PC for $\hat{\Lambda}\hat{\Lambda}^\top$ vs $\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$. The gap is the uniqueness $\hat{\Psi}_a$.

New in this version: posterior CIs on the loadings are now available from draws-level PCA. The figure below propagates the Λ posterior to give 90% marginal CIs on each rotated loading. A feature is **reliable** (solid) if its CI excludes zero.

Fixed-reference PCA (methods). The eigenvectors of a per-draw $\Lambda_s\Lambda_s^\top$ can rotate freely across posterior draws — the leading two- or three-dimensional subspace is identified by the data, but its internal orientation is not. Per supervisor guidance, we therefore fix the reference orientation

to the eigendecomposition of the posterior mean $\bar{\Lambda}\bar{\Lambda}^\top$ (SVD $\rightarrow V_{\text{ref}}$, sign-anchored via the largest-magnitude entry per column), and each draw's loading vector is projected onto this fixed V_{ref} before the 90% marginal CI is formed (`scripts/31_pca_with_ci.R`). This is a first-order (mean-rotation) approximation of a fully Procrustes-aligned per-draw CI; the *reliable/unreliable* flag we report is robust to the approximation because it depends on the CI's *sign*, not its width.

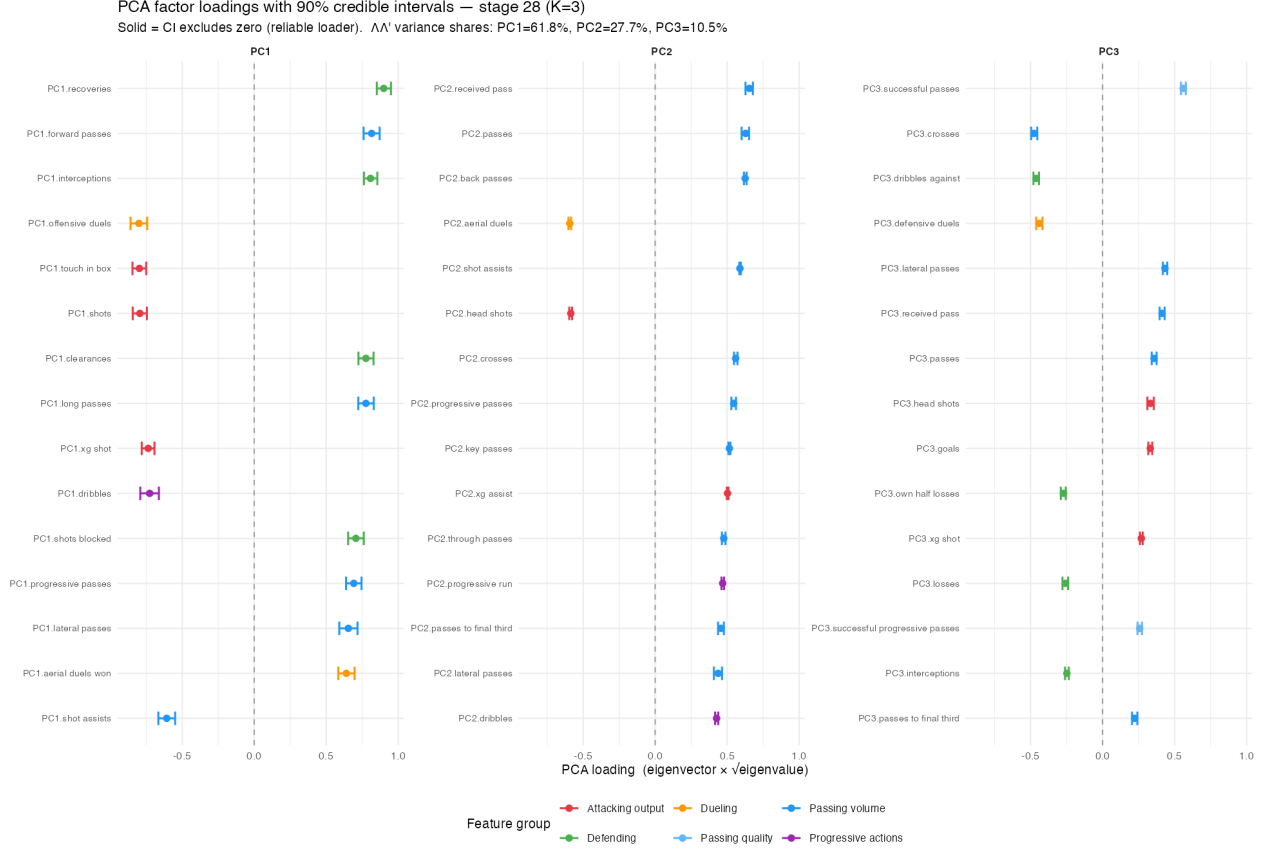


Figure 5: PCA factor loadings with 90% credible intervals (K=3, $P = 48$). Solid = CI excludes zero. PC1: defensive work-rate (positive) vs attacking output (negative). PC2: pass-based / technical (positive) vs aerial / physical (negative). PC3: possession quality (positive) vs wide / direct play (negative).

Table 6: Top 5 reliable loaders per PC (CI excludes zero). Positive PC1 = defensive; negative PC1 = attacking. PC2: positive = technical/pass; negative = aerial/physical. PC3: positive = possession quality; negative = wide/direct.

PC	Feature	Group	Loading	CI lo	CI hi
PC1	recoveries	Defending	0.900	0.852	0.950
PC1	forward passes	Passing volume	0.816	0.760	0.871
PC1	interceptions	Defending	0.808	0.762	0.855
PC1	offensive duels	Dueling	-0.801	-0.859	-0.743
PC1	touch in box	Attacking output	-0.798	-0.846	-0.750
PC2	received pass	Passing volume	0.655	0.627	0.679

PC	Feature	Group	Loading	CI lo	CI hi
PC2	passes	Passing volume	0.628	0.600	0.652
PC2	back passes	Passing volume	0.625	0.616	0.635
PC2	aerial duels	Dueling	-0.593	-0.602	-0.584
PC2	shot assists	Passing volume	0.589	0.585	0.595
PC3	successful passes	Passing quality	0.560	0.545	0.578
PC3	crosses	Passing volume	-0.476	-0.496	-0.454
PC3	dribbles against	Defending	-0.461	-0.480	-0.442
PC3	defensive duels	Dueling	-0.439	-0.461	-0.418
PC3	lateral passes	Passing volume	0.433	0.419	0.449

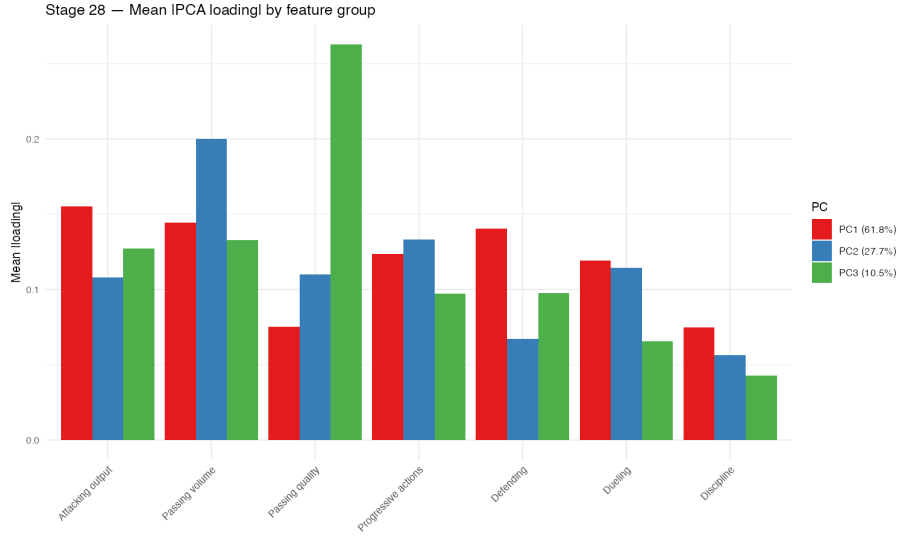


Figure 6: Mean absolute PCA loading by feature group and PC (K=3). Visualises which skill domains define each factor direction.

3.3 Player archetypes

Projecting posterior-mean player effects onto the PCA rotation gives each player’s archetype position. K-means clustering ($k = 5$) in PC1–PC2 space identifies five archetypes, ordered by PC1: Forwards (high attacking) through Centre backs (high defensive).

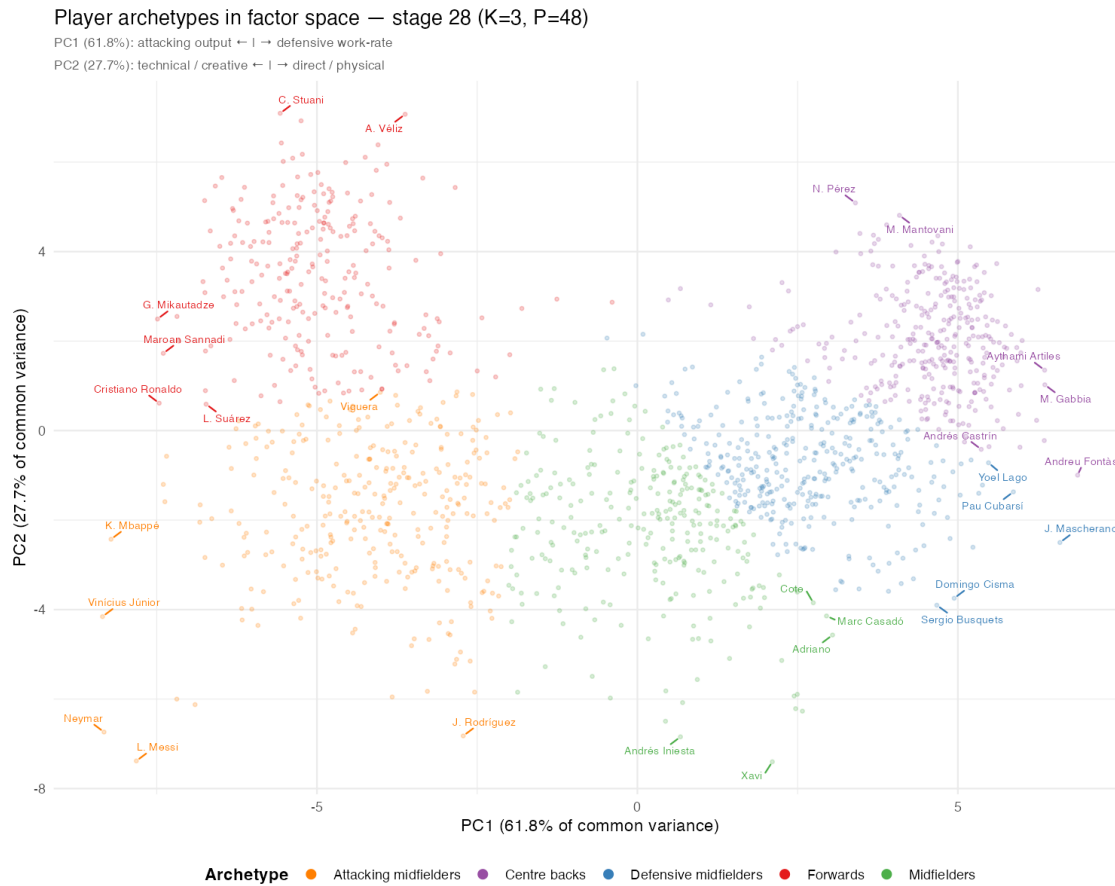


Figure 7: Player archetypes: PC1 vs PC2 for 1,529 outfield players, coloured by k-means cluster (k=5). PC1 negative = high attacking output; PC1 positive = high defensive work-rate. PC2 separates technical/creative from direct/physical profiles.

Stage 28 — PC1 archetype players (61.8% of shared variance)

Highest PC1	Lowest PC1
1. Andreu Fontàs 6.869 (4s)	1. Vinicius Júnior -8.346 (8s)
2. J. Mascherano 6.592 (4s)	2. Neymar -8.322 (3s)
3. M. Gabbia 6.357 (1s)	3. K. Mbappé -8.213 (2s)
4. Aythami Ariles 6.354 (2s)	4. L. Messi -7.818 (7s)
5. Zé Castro 6.349 (2s)	5. G. Mikautadze -7.488 (1s)
6. Antonio Amaya 6.251 (2s)	6. Cristiano Ronaldo -7.461 (4s)
7. Gerard Piqué 6.228 (8s)	7. Maroan Sannadi -7.398 (1s)
8. A. Ba 6.003 (2s)	8. N. Amrabat -7.397 (3s)
9. Z. Feddal 5.988 (5s)	9. Alfon González -7.373 (1s)
10. Víctor Ruiz 5.983 (8s)	10. M. Depay -7.346 (2s)
11. S. Umtiti 5.969 (5s)	11. Lamine Yamal -7.187 (3s)
12. W. Hoedt 5.945 (1s)	12. A. Isak -7.183 (3s)
13. G. Cabral 5.924 (5s)	13. K. Toko-Ekambi -7.155 (2s)
14. NA 5.919 (1s)	14. O. Dembélé -6.902 (5s)
15. Pau Cubarsi 5.866 (3s)	15. M. Cho -6.828 (1s)

Figure 8: Top and bottom 15 players on PC1 (Defensive work-rate vs Attacking output).

Stage 28 — PC2 archetype players (27.7% of shared variance)

Highest PC2	Lowest PC2
1. C. Stuani 7.091 (6s)	1. Xavi -7.401 (1s)
2. A. Véliz 7.069 (1s)	2. L. Messi -7.382 (7s)
3. Chris Ramos 6.922 (2s)	3. Andrés Iniesta -6.844 (4s)
4. A. Budimir 6.427 (7s)	4. J. Rodríguez -6.822 (3s)
5. Manucho Gonçalves 6.385 (2s)	5. Neymar -6.737 (3s)
6. E. Boateng 6.175 (2s)	6. É. Banega -6.495 (5s)
7. S. Okazaki 6.108 (1s)	7. Dani Alves -6.270 (3s)
8. Y. En-Nesyri 6.087 (7s)	8. T. Kroos -6.214 (10s)
9. Juanmi Latasa 6.015 (3s)	9. O. Dembélé -6.124 (5s)
10. B. Yıldırım 5.990 (1s)	10. Marcelo -6.078 (8s)
11. Toché 5.951 (1s)	11. Lamine Yamal -6.000 (3s)
12. G. Carrillo 5.816 (4s)	12. Pedri -5.977 (6s)
13. Jon Bautista 5.767 (1s)	13. Jonathan Viera -5.953 (4s)
14. S. Weissman 5.724 (2s)	14. T. Gayoso Roberto -5.932 (2s)
15. Diego García 5.669 (1s)	15. M. Pjanić -5.893 (1s)

Figure 9: Top and bottom 15 players on PC2 (Technical/pass-based vs Aerial/physical).

Stage 28 — PC3 archetype players (10.5% of shared variance)

Highest PC3	Lowest PC3
1. Iván Alejo 5.004 (5s)	1. L. Messi -6.066 (7s)
2. Juan Iglesias 4.550 (6s)	2. Xavi -5.822 (1s)
3. O. Vranješ 4.456 (1s)	3. M. Pjanić -5.464 (1s)
4. Houboulang Mendes 4.203 (1s)	4. Cristiano Ronaldo -4.920 (4s)
5. A. Aznou 4.099 (1s)	5. K. Benzema -4.523 (9s)
6. Miguel Llambrich 3.973 (1s)	6. T. Kroos -4.131 (10s)
7. Rubén Sánchez 3.970 (1s)	7. Arthur -3.959 (3s)
8. Davinchi 3.851 (1s)	8. Andrés Iniesta -3.948 (4s)
9. Quini 3.844 (5s)	9. I. Gündoğan -3.928 (1s)
10. L. Blanco 3.770 (1s)	10. Paulinho -3.909 (1s)
11. Pacha 3.727 (5s)	11. Pedro -3.867 (1s)
12. Jesús Vázquez 3.569 (4s)	12. Pau Cubarsí -3.836 (3s)
13. S. Dubarbier 3.558 (1s)	13. A. Christensen -3.721 (3s)
14. Pedro Porro 3.546 (2s)	14. S. Nasri -3.703 (1s)
15. Aitor Buñuel 3.536 (1s)	15. Philippe Coutinho -3.657 (4s)

Figure 10: Top and bottom 15 players on PC3 (Possession quality vs Wide/direct play).

3.4 Player similarity — the short version

i Note

For the meeting. This box is a two-minute summary of the similarity result. The full technical build-up (percentile radars, the classical distance metric, the fully Bayesian version, and the population reliability check) follows below in §3.5 and its “Style vs. level” subsection.

The question. What does it mean for two football players to be “similar”? Two players can look alike because they produce similar *numbers* (same goals, same passes), or because they play the same *way* even if one produces much more than the other. Our model lets us tell these two apart.

Step 1 — what “similar” looks like. The figure below compares two players on eight football-skill groups (attack, shooting, passing, progression, defending, dueling, discipline), shown as a percentile against all 1,529 outfield players in the dataset. Messi and Neymar’s shapes are nearly identical — that is what a genuinely similar pair looks like under this model.

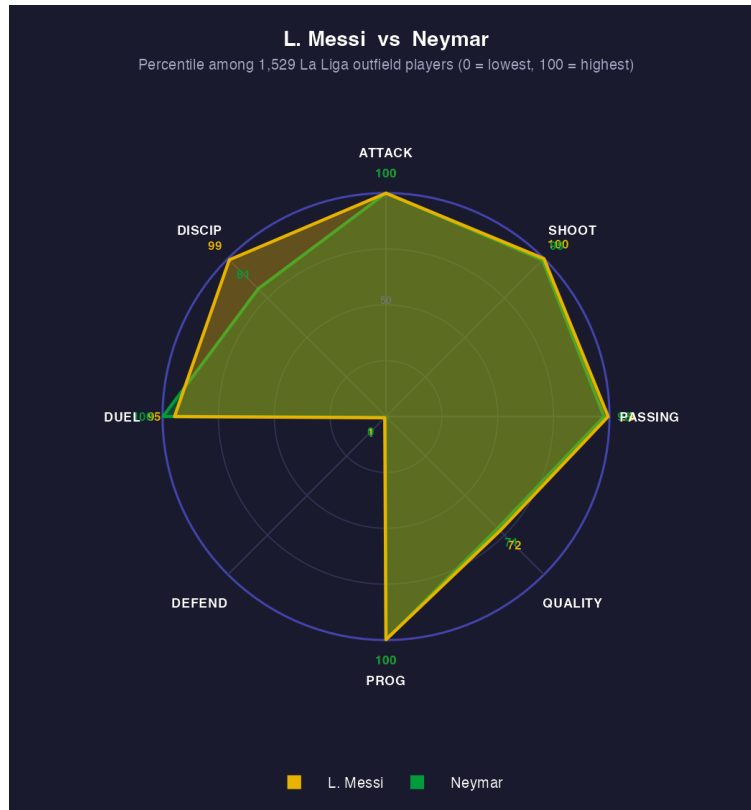


Figure 11: Messi vs Neymar: the closest pair of players in the whole dataset. Nearly identical shape across all eight skill groups.

Step 2 — separating “how they play” from “how much they produce.” The model splits each player’s profile into a *shared style* part (their position relative to the three main stylistic axes the model finds in the data) and a *personal, unshared* part (whatever makes that specific player over- or under-perform on specific numbers like goals, beyond what their style predicts). Two players can be close on style while being far apart on that personal part — the figure below shows this for a broad set of players: points toward the bottom-right are players who play alike but produce very differently.

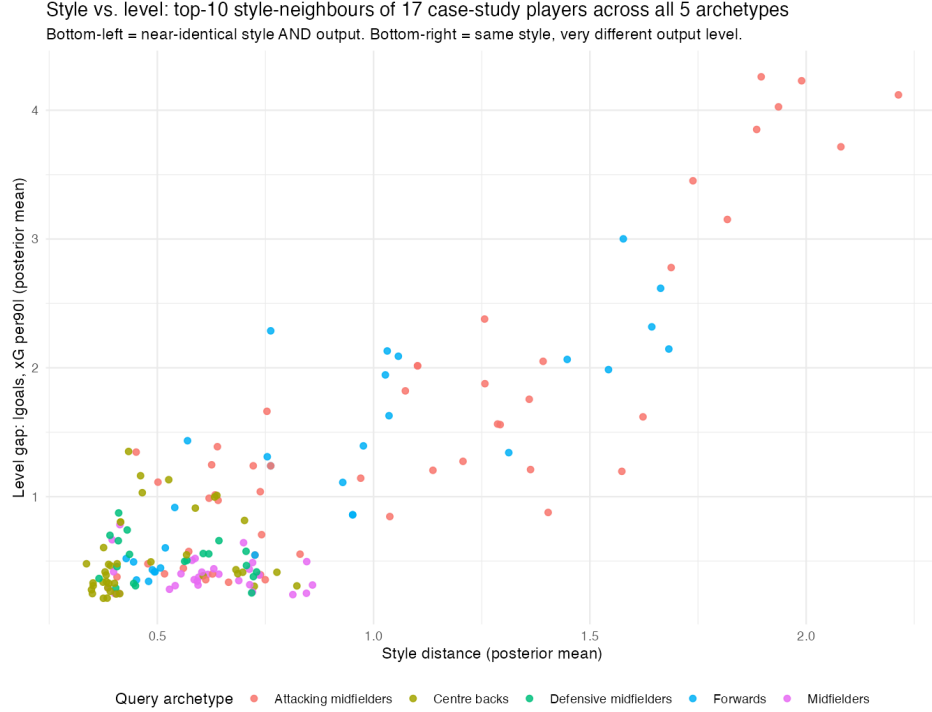


Figure 12: Style distance (how alike two players play) vs. level gap (how differently they produce), for the ten most similar players to each of 17 case-study players. Bottom-right = same style, different output tier.

Step 3 — why “the closest player” needs a confidence level, not just a number. Because the model is Bayesian, a player’s position is not a single dot — it is a cloud of plausible positions. The figure below shows this directly for two cases already discussed above: Messi and Neymar’s clouds overlap almost completely (we can say with real confidence that they are each other’s closest match), while four elite centre-backs’ clouds overlap messily with each other (the “closest match” among them is much closer to a coin flip than a confident answer). A plain distance number cannot tell these two situations apart; the posterior can.

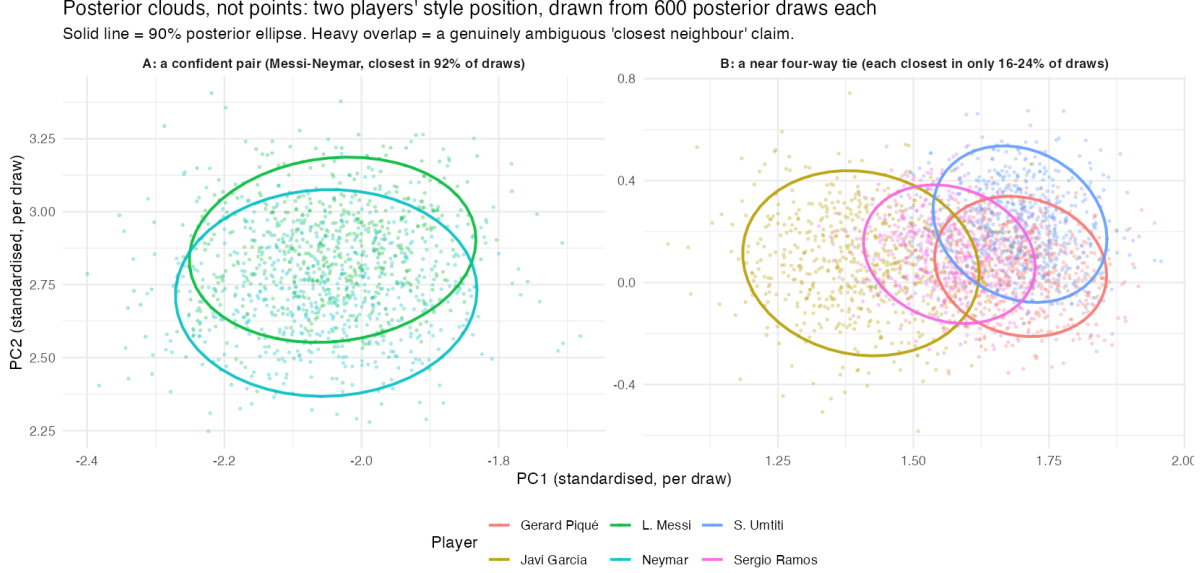


Figure 13: Two players are not two dots: each has a cloud of plausible positions. Left: Messi and Neymar’s clouds overlap almost completely (closest match 92% of the time). Right: four centre-backs whose clouds overlap messily (each one is the ‘closest’ only 16-24% of the time) – a genuine four-way tie, not a confident single answer.

Bottom line. A single “most similar player” number can hide two very different situations: a genuinely confident match, or a coin-flip among several candidates that happens to produce one arbitrary “winner.” This model tells the two apart, and additionally separates *style* from *output level* — which is what makes it useful for scouting: finding a player who plays like a target but at a different (typically cheaper) output tier, rather than pretending a replacement for an outlier talent exists.

3.5 Player profiles and comparisons

3.5.1 Construction

Step 1 — Player effects. The starting point is $\hat{a}_{i,p}$, the posterior mean player effect for player i and feature p . These are extracted directly from the fitted model: because the model decomposes each observation as $y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n$, the player effect has already had season-wide shifts (b_j) and residual noise (ε_n) removed. What remains is each player’s stable latent identity in the original 48-dimensional Z-scored feature space — not a dimension-reduced projection.

Step 2 — Loading-weighted group composite. The 48 features are assigned to eight groups (Table below). For each group g and player i :

$$\text{composite}_{i,g} = \frac{\sum_{p \in g} |\lambda_{p,1}| \hat{a}_{i,p}}{\sum_{p \in g} |\lambda_{p,1}|}$$

where $\lambda_{p,1}$ is the PC1 loading of feature p from the PCA of $\hat{\Lambda}\hat{\Lambda}^\top$ (§3.2). Using absolute loading magnitude as weight gives more influence to features that explain more inter-player covariance in the factor model — connecting the radar directly to the model’s structure. Features absent from the loading table fall back to weight 1.

One practical adjustment was required: the two original “Shooting rates” features (`rate_shots_on_target`, `rate_goal_conversion`) were dropped from the model in the feature-selection step (low ICC). The SHOOT spoke is therefore populated by reassigning the finishing features from the Attacking output group — `per90_goals`, `per90_shots`, `per90_xg_shot`, `per90_head_shots` — while the ATTACK spoke retains the creation and box-presence features (`per90_touch_in_box`, `per90_assists`, `per90_xg_assist`). This gives a more informative split: ATTACK measures how much a player creates chances; SHOOT measures how much they finish them.

Step 3 — Percentile conversion. Each composite is ranked among all 1,529 outfield players and converted to a 0–100 scale:

$$\text{pct}_{i,g} = \frac{\text{rank}(\text{composite}_{i,g})}{1,529} \times 100$$

A score of 90 means the player’s loading-weighted effect in group g exceeds 90% of the La Liga outfield pool. 50 is the median player; the scale is the same for all pairs.

What this is not. The spoke values are not PC scores (those appear only in the scatter plot and similarity metric), not raw statistics, and not ICC values. The radars show a player’s purified, season-adjusted group fingerprint on the model’s own scale, converted to a percentile for legibility.

Direction correction. For features where a higher raw value is worse, the sign is flipped before computing the composite, so that 100th percentile always means “better”:

- *Discipline:* `per90_fouls` and `per90_yellow_cards` are negated (fewer = better). `per90_fouls_suffered` is excluded from this spoke entirely — being fouled is position-dependent (attackers are fouled far more than defenders) and does not reflect disciplinary behaviour.
- *Defending:* `per90_losses`, `per90_own_half_losses`, and `per90_dangerous_own_half_losses` are negated (fewer ball losses = better defending).

All other features are kept as-is; higher Z-score already corresponds to more of a desirable quantity (more recoveries, more dribbles, higher pass completion, etc.).

Table 7: Eight radar spokes, their group description, and representative features.

Spoke	Description	Key features (examples)
AT-TACK	Attacking output	touches in box, assists, xG assist
SHOOT	Shooting / finishing	goals, shots, xG per shot, headed shots
PASS-ING	Passing volume	passes, progressive passes, crosses, smart passes
QUAL-ITY	Passing quality	pass completion rate, progressive pass completion
PROG	Progressive actions	dribbles, progressive runs, accelerations

Spoke	Description	Key features (examples)
DE-FEND	Defending	recoveries, interceptions, clearances, sliding tackles
DUEL	Dueling	defensive/offensive/aerial/pressing duels
DISCIP	Discipline	fouls committed (negated), yellow cards (negated)

3.5.2 Pair comparisons

Each radar shows a player’s percentile among 1,529 La Liga outfield players; higher = larger loading-weighted player effect relative to the pool. All six players below appeared in La Liga during 2011–2023.

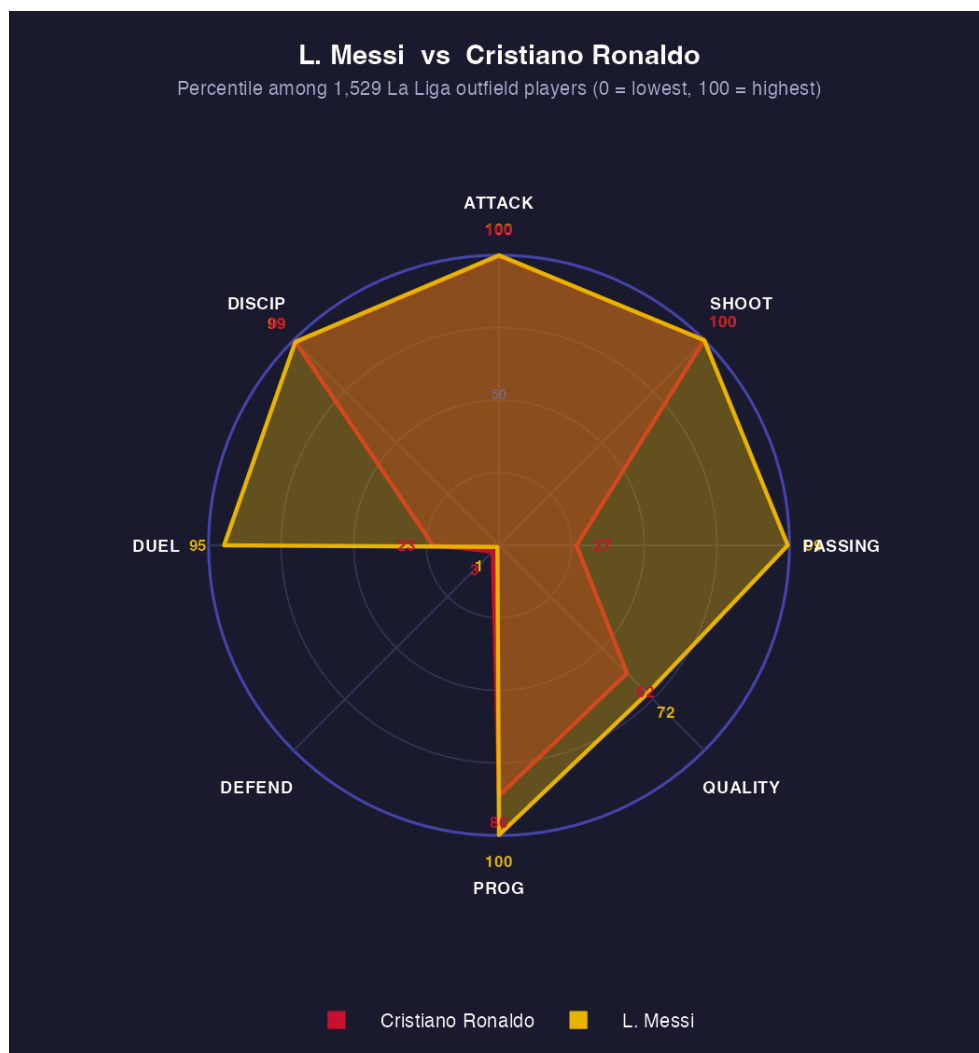


Figure 14: Messi vs Cristiano Ronaldo. Cristiano is the top SHOOT scorer in the dataset (100th percentile) but low on PASSING (30th). Messi is elite across all spokes, with the strongest edge in Passing quality and Progressive actions.

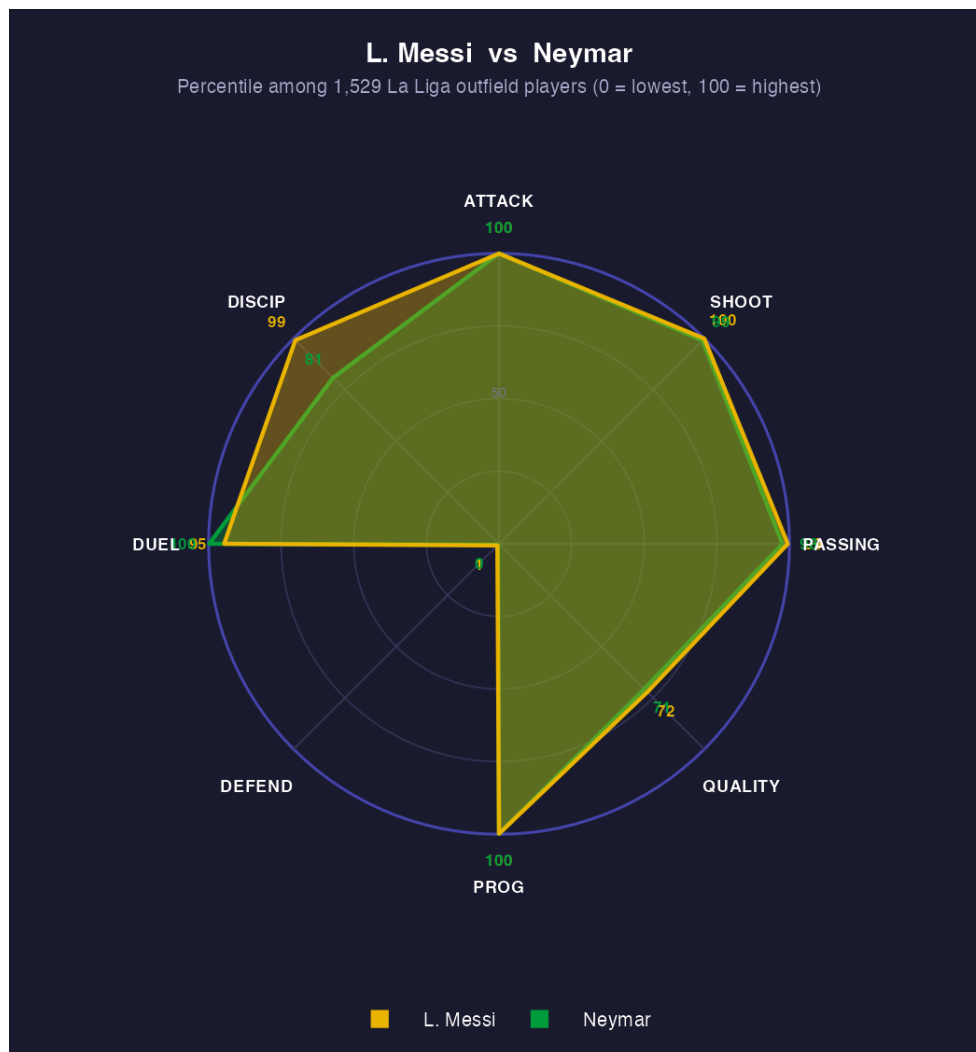


Figure 15: Messi vs Neymar — the closest pair in archetype space ($d = 1.74$, similarity rank 1). The two profiles are nearly identical; the visible separation is in Shooting and Discipline.

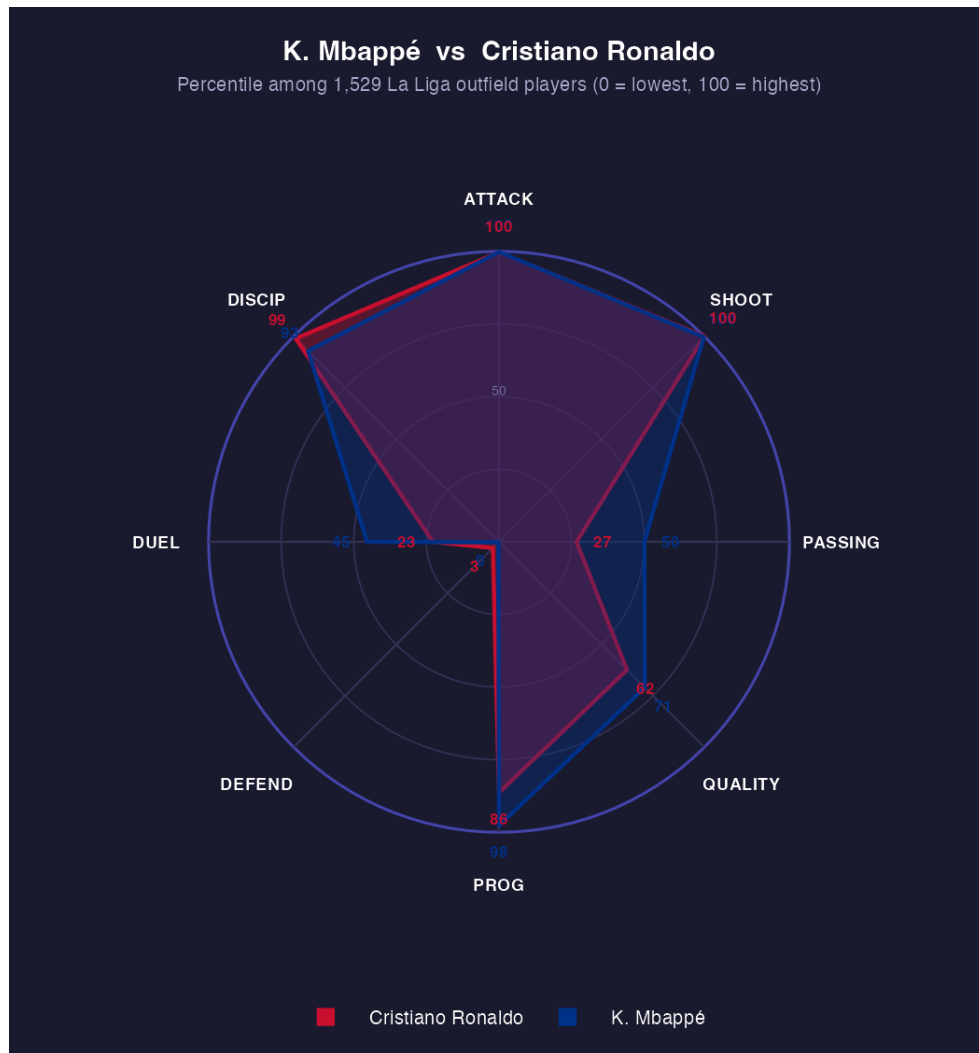


Figure 16: Mbappé vs Cristiano Ronaldo: two elite finishers from different eras. Both score near 100 on SHOOT; Mbappé shows a clear edge in Progressive actions (dribbling, acceleration) while Cristiano leads in Dueling.

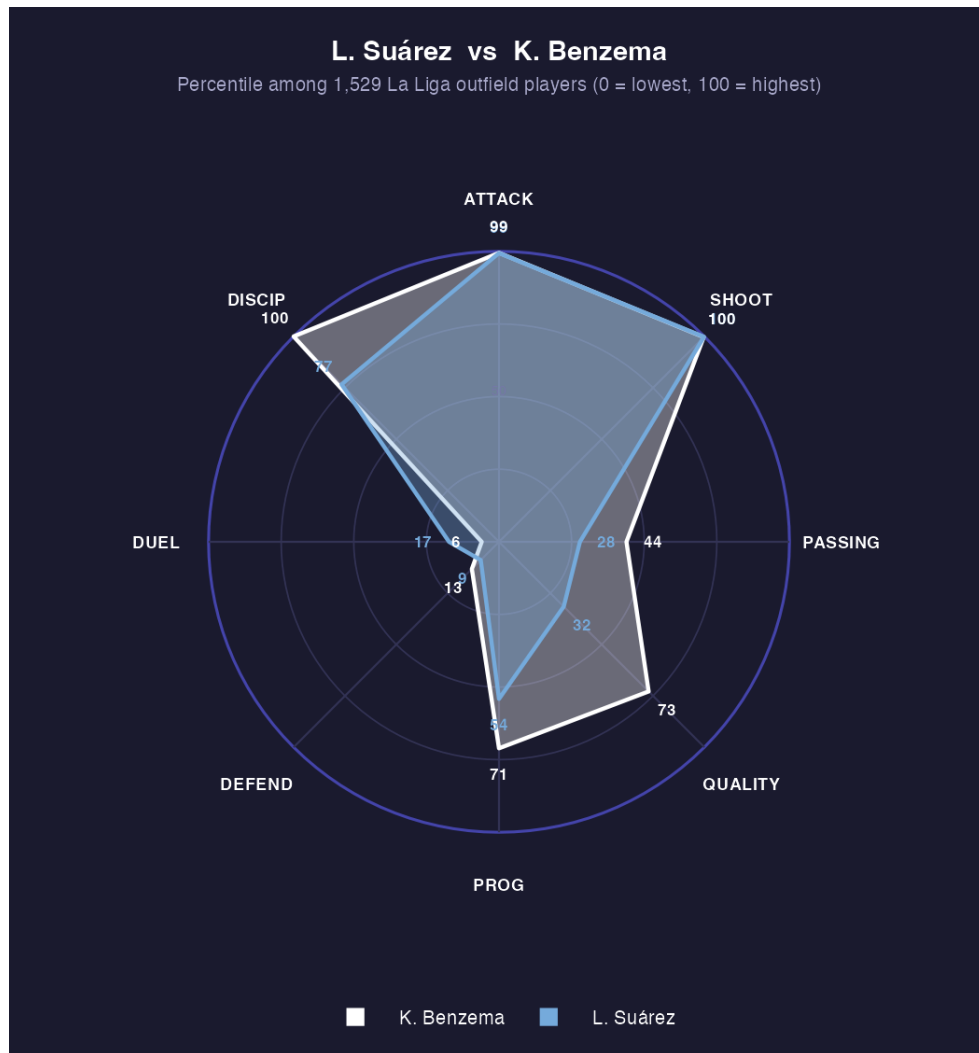


Figure 17: Suárez vs Benzema: two elite centre forwards with contrasting styles. Suárez scores higher in Dueling; Benzema leads in Passing and Attacking output, reflecting his role as a link player.

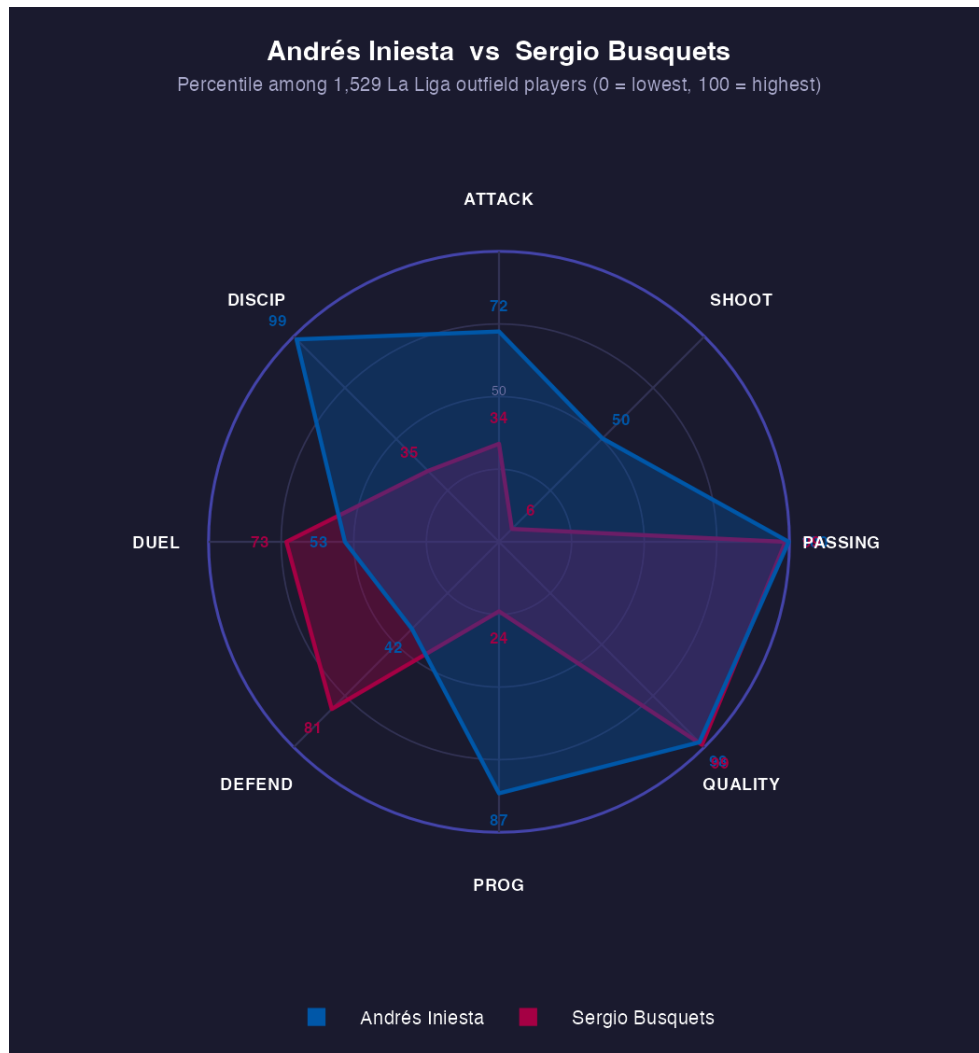


Figure 18: Iniesta vs Busquets: contrasting central midfielder profiles from the same team. Iniesta leads in Passing quality and Attacking output; Busquets leads in Defending and Dueling.

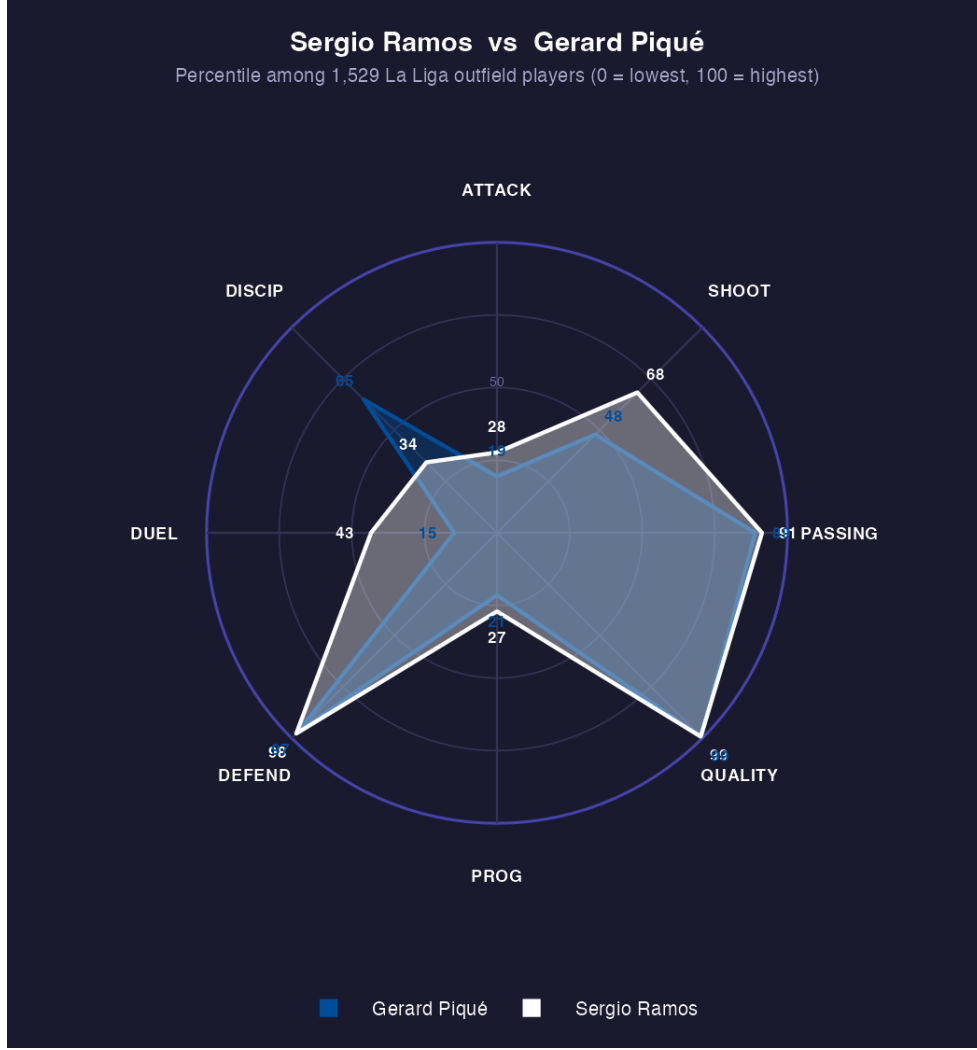


Figure 19: Ramos vs Piqué: two elite centre backs. Ramos leads in Dueling; Piqué shows higher Passing quality — consistent with their well-known playing styles.

3.5.3 Player similarity

The radar compares players within a single group at a time. For a global similarity score, we use a different (and complementary) representation.

Each player's posterior-mean effect $\hat{a}_i \in \mathbb{R}^{48}$ is projected onto the three PCA eigenvectors of $\hat{\Lambda}\hat{\Lambda}^\top$, giving scores $(s_{i,1}, s_{i,2}, s_{i,3})$ that summarise the player's archetype in a three-number summary. These are the PC scores shown in the scatter plot. Each PC is then standardised to unit variance across the 1,529 players before computing distance, so all three dimensions contribute equally regardless of their raw scale. The similarity between players i and i' is:

$$d(i, i') = \left\| \begin{pmatrix} \tilde{s}_{i,1} \\ \tilde{s}_{i,2} \\ \tilde{s}_{i,3} \end{pmatrix} - \begin{pmatrix} \tilde{s}_{i',1} \\ \tilde{s}_{i',2} \\ \tilde{s}_{i',3} \end{pmatrix} \right\|_2, \quad \tilde{s}_{i,k} = \frac{s_{i,k} - \bar{s}_k}{\text{sd}(s_k)}$$

A distance of 1 unit $\approx$ one SD of combined-archetype separation across the pool.

Table 8: Top-5 nearest neighbours in archetype space for four named players. Messi’s nearest neighbour is Neymar (dist,=,1.74); Ramos’s is Piqué (dist,=,0.16).

Player	Rank	Similar player	Distance
Sergio Ramos	1	Javi García	0.161
Sergio Ramos	2	Gerard Piqué	0.164
Sergio Ramos	3	S. Umtiti	0.316
Sergio Ramos	4	A. Christensen	0.382
Sergio Ramos	5	A. Rüdiger	0.424
K. Benzema	1	Cristiano Ronaldo	0.562
K. Benzema	2	S. Jovetić	0.774
K. Benzema	3	L. Suárez	0.827
K. Benzema	4	Rubén Castro	1.186
K. Benzema	5	K. Mbappé	1.193
Sergio Busquets	1	Damián Rodríguez	0.454
Sergio Busquets	2	Marc Casadó	0.476
Sergio Busquets	3	Z. Kuzmanović	0.570
Sergio Busquets	4	M. Díaz	0.594
Sergio Busquets	5	Dean Huijsen	0.594
L. Messi	1	Neymar	1.736
L. Messi	2	Pedro	2.014
L. Messi	3	Philippe Coutinho	2.115
L. Messi	4	J. Rodríguez	2.155
L. Messi	5	Jonathan Viera	2.181

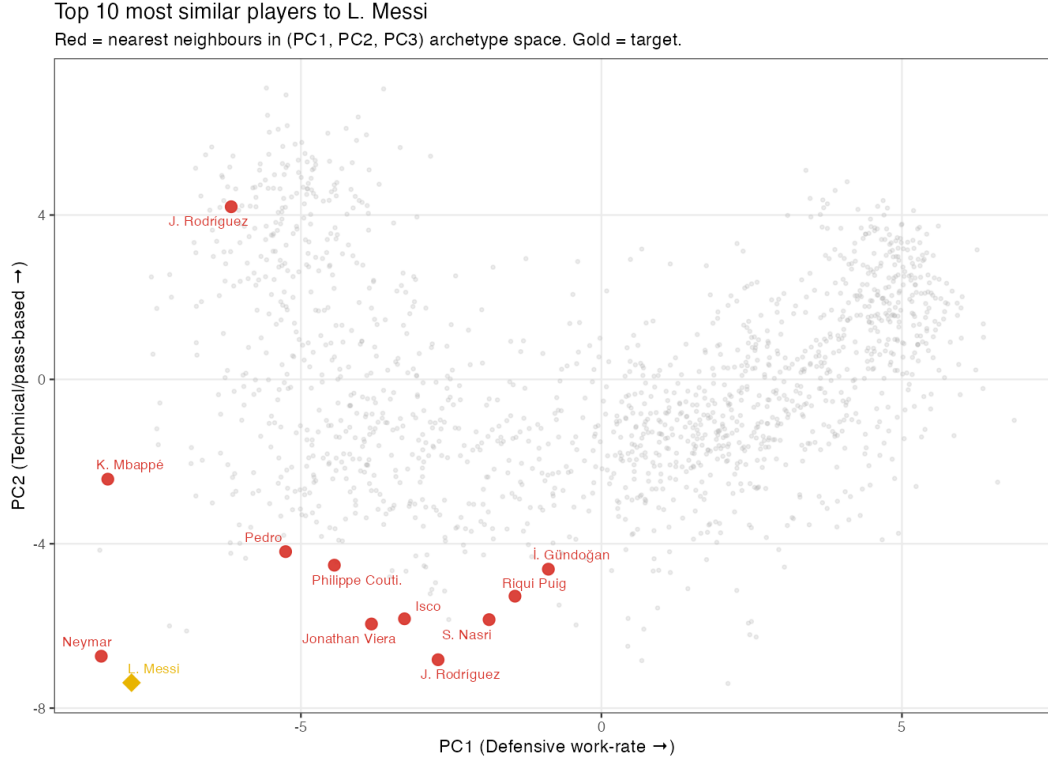


Figure 20: Top 10 most similar players to L. Messi on the $PC1 \times PC2$ scatter. Gold diamond = Messi; red = his 10 nearest neighbours; grey = all other players. Neighbours are Neymar, Pedro, Coutinho, and other creative attackers.

3.5.4 Style vs. level: a fully Bayesian similarity score

The similarity metric above is a useful first pass, but it is a **point estimate** on **classical** PCA scores, and it blends two conceptually different things into one number: whether two players *look alike* (the shape of their profile) and whether they *produce alike* (their level of output). This matters for a natural question the point-estimate metric cannot answer on its own: can a player be genuinely similar in style to an exceptional outlier — a generational talent — without matching their output? Intuitively, yes: two players can share a nearly identical profile shape while sitting at very different output levels, for reasons the model does not (and should not) collapse into a single “similarity” number — squad role, finishing variance, luck, and the sheer rarity of truly elite output all separate “style” from “level.”

The model’s own covariance structure already gives a principled way to make this split. Recall (§1) that the player effect decomposes as $\mathbf{a}_i = \Lambda \eta_i + \psi_a \odot \mathbf{z}_{u,i}$: a **shared** part driven by the $K = 3$ factors, and an **idiosyncratic** part that the shared factors do not explain. “Style” is naturally the player’s position in the shared factor space — the PCA-rotated score $\mathbf{f}_i$ used throughout this report — while “level” is naturally read off the specific output features the shared style position does not by itself pin down (goals and xG per shot). Two players can be **style-close** (nearly identical $\mathbf{f}_i$) while being **level-far** (very different modelled goals/xG) — that is the formal, model-native version of “similar style, different output.”

This section recomputes the archetype-space distance **per posterior draw** (not once on posterior means), for the case-study players already introduced above plus their recovered nearest neighbours. Each pair below therefore carries a genuine credible interval on the style distance and on the level gap, plus a stability statement: the fraction of posterior draws in which a given neighbour is *the* single closest player in style space. A tight interval and a high stability fraction mean the ranking is not an artefact of posterior noise; a wide interval or low stability means the ordering among the closest few neighbours should be read loosely.

Table 9: Top-3 style-neighbours per case-study player, fully Bayesian version: posterior mean style distance and level gap (goals + xG per90, Z-scored units), each with a 90% credible interval, plus the fraction of posterior draws in which the neighbour is the single closest player in style space.

Player	Style neighbour	Style dist. (90% CI)	P(closest)	Level gap (90% CI)
Cristiano Ronaldo	K. Benzema	0.76 [0.37, 1.22]	61%	1.24 [0.71, 1.75]
Cristiano Ronaldo	L. Suárez	0.95 [0.45, 1.48]	20%	0.86 [0.37, 1.32]
Cristiano Ronaldo	S. Jovetić	1.06 [0.49, 1.70]	16%	2.09 [1.39, 2.81]
Gerard Piqué	S. Umtiti	0.38 [0.13, 0.72]	41%	0.60 [0.32, 0.89]
Gerard Piqué	Sergio Ramos	0.42 [0.15, 0.82]	31%	0.80 [0.45, 1.16]
Gerard Piqué	Javi García	0.49 [0.20, 0.82]	16%	0.49 [0.17, 0.84]
K. Benzema	S. Jovetić	0.64 [0.23, 1.17]	39%	0.97 [0.41, 1.57]
K. Benzema	L. Suárez	0.73 [0.47, 1.03]	11%	0.55 [0.17, 0.95]
K. Benzema	Cristiano Ronaldo	0.76 [0.37, 1.22]	25%	1.24 [0.71, 1.75]
L. Messi	Neymar	1.10 [0.60, 1.64]	92%	2.02 [1.44, 2.59]
L. Messi	Pedro	1.69 [1.26, 2.13]	5%	2.78 [2.14, 3.45]
L. Messi	J. Rodríguez	1.74 [1.31, 2.20]	1%	3.45 [2.92, 3.98]
L. Suárez	S. Jovetić	0.57 [0.23, 0.99]	40%	1.43 [0.86, 2.04]
L. Suárez	K. Benzema	0.73 [0.47, 1.03]	10%	0.55 [0.17, 0.95]
L. Suárez	M. Depay	0.75 [0.28, 1.30]	18%	1.31 [0.79, 1.87]
Neymar	L. Messi	1.10 [0.60, 1.64]	46%	2.02 [1.44, 2.59]
Neymar	Jonathan Viera	1.26 [0.96, 1.58]	7%	1.88 [1.34, 2.40]
Neymar	J. Rodríguez	1.29 [1.03, 1.57]	5%	1.56 [1.03, 2.09]
Sergio Ramos	Gerard Piqué	0.42 [0.15, 0.82]	24%	0.80 [0.45, 1.16]
Sergio Ramos	S. Umtiti	0.43 [0.18, 0.77]	16%	1.35 [0.97, 1.74]
Sergio Ramos	Javi García	0.46 [0.16, 0.88]	17%	1.16 [0.70, 1.61]

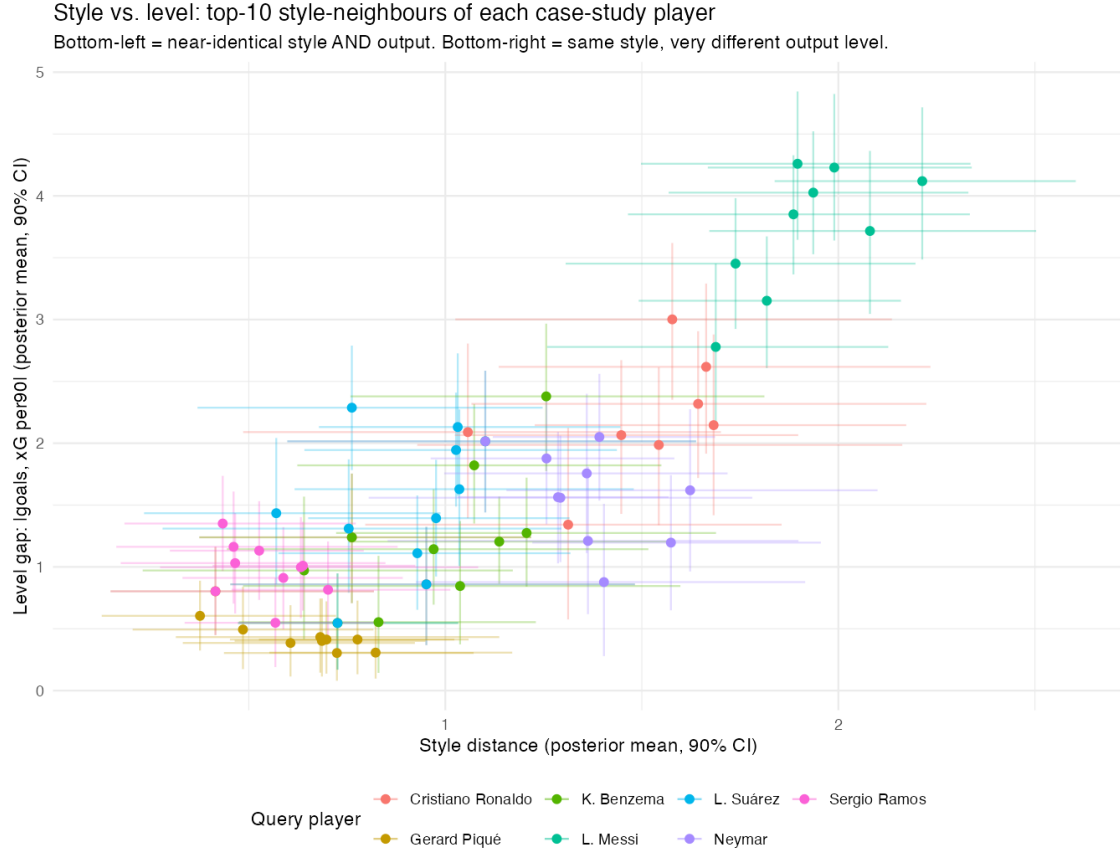


Figure 21: Style distance vs. level gap for the top-10 style-neighbours of each case-study player. Points toward the bottom-right are the clearest examples of ‘same shape, different output level’ — a stylistic twin at a different tier. Error bars are 90% credible intervals from the posterior draws.

Two concrete patterns emerge from this table that were not visible under the earlier point-estimate metric. **Benzema’s closest style-neighbour is not Cristiano Ronaldo.** The point-estimate radar metric ranked Ronaldo first (they were team-mates, and the percentile radar’s Attacking-output spoke blends shape and level together); the posterior-draw style-only metric instead makes S. Jovetić the modal closest match (39% of draws) with Ronaldo a clear second (25%) — and, more to the point, Ronaldo also carries the *largest* level gap of Benzema’s four closest style-neighbours. Read together, this is a direct, data-driven instance of the argument above: even Benzema, an elite striker whose *shape* is genuinely close to Ronaldo’s, cannot close the *level* gap to him — Ronaldo’s idiosyncratic scoring output separates him from his closest stylistic peers, not just from the pool at large. **Ramos’s closest neighbour is a near four-way tie, not a clean winner.** The old point estimate placed Javi García first by a margin of 0.003 over Gerard Piqué (0.1611 vs. 0.1641) — a difference with no meaningful posterior support. The draw-level treatment shows four centre-backs (Piqué, Umtiti, Javi García, Mathieu) within a similar style-distance range, each closest-neighbour in only 16–24% of draws: Piqué is the modal pick, consistent with the well-known Ramos–Piqué pairing used earlier in this report, but the honest statement is a four-way toss-up, not a single confident answer. This is the practical payoff of moving to a fully Bayesian metric: a point estimate cannot distinguish “clearly the closest neighbour” from “one of several statistically indistinguishable candidates,” and

here the two cases look identical until the posterior draws are examined individually.

Two things follow from this decomposition. First, it explains *why* a true generational outlier is unlikely to have a perfect twin: the outlier-ness lives disproportionately in the idiosyncratic term $\psi_a \odot \mathbf{z}_{u,i}$, which is by construction the part of a player’s identity that is *not* shared with anyone else — so matching someone’s shared style is achievable, but matching both their style and their most idiosyncratic output is structurally much less likely. (The dataset is La Liga only, 2011/12–2022/23; a player like Haaland has never played in this league and is not in this analysis — he is a useful illustration of the “generational outlier” idea, not a literal query target. The case-study players above, all of whom did play in this window, serve as the concrete stand-ins.) Second, and more usefully, it turns “style-close, level-far” pairs into a concrete scouting/recommendation product: a shortlist of players who match a target’s profile shape at a different output tier — a candidate list of plausible, cheaper alternatives, rather than attempted replacements for an irreplaceable outlier. As with the point-estimate similarity metric above, this is a style/production tool only: market value, age, injury history, and league context are not modelled, and any shortlist it produces should be read as a starting point for scouting judgement, not a substitute for it.

3.6 Season profiles

Season effects $\hat{b}_j$ capture year-specific league-wide shifts. ICC_season < 5% for most features (season explains little of total variance), but group-averaged trends reveal coherent tactical changes across La Liga’s 12 seasons.

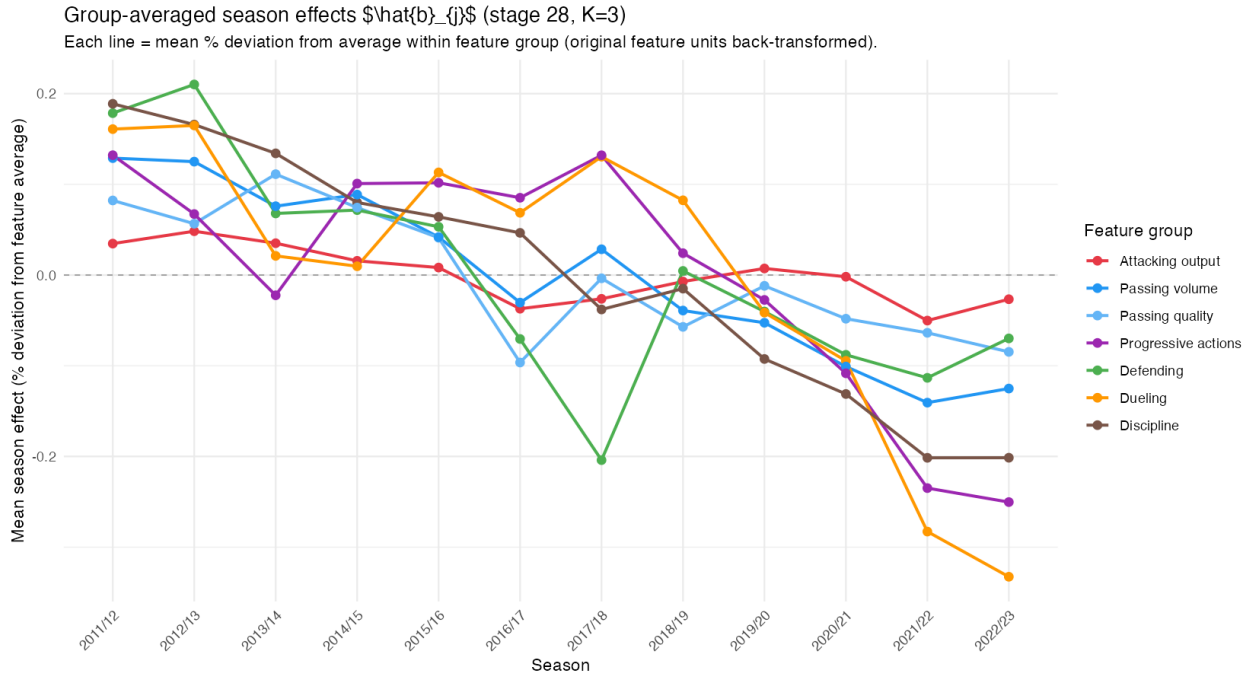


Figure 22: Group-averaged season effects: posterior mean $\hat{b}_{j,p}$ averaged within each feature group. Ribbons = mean 90% CIs across features in the group. Reveals league-wide tactical shifts (K=3 t-model, 2011/12–2022/23).

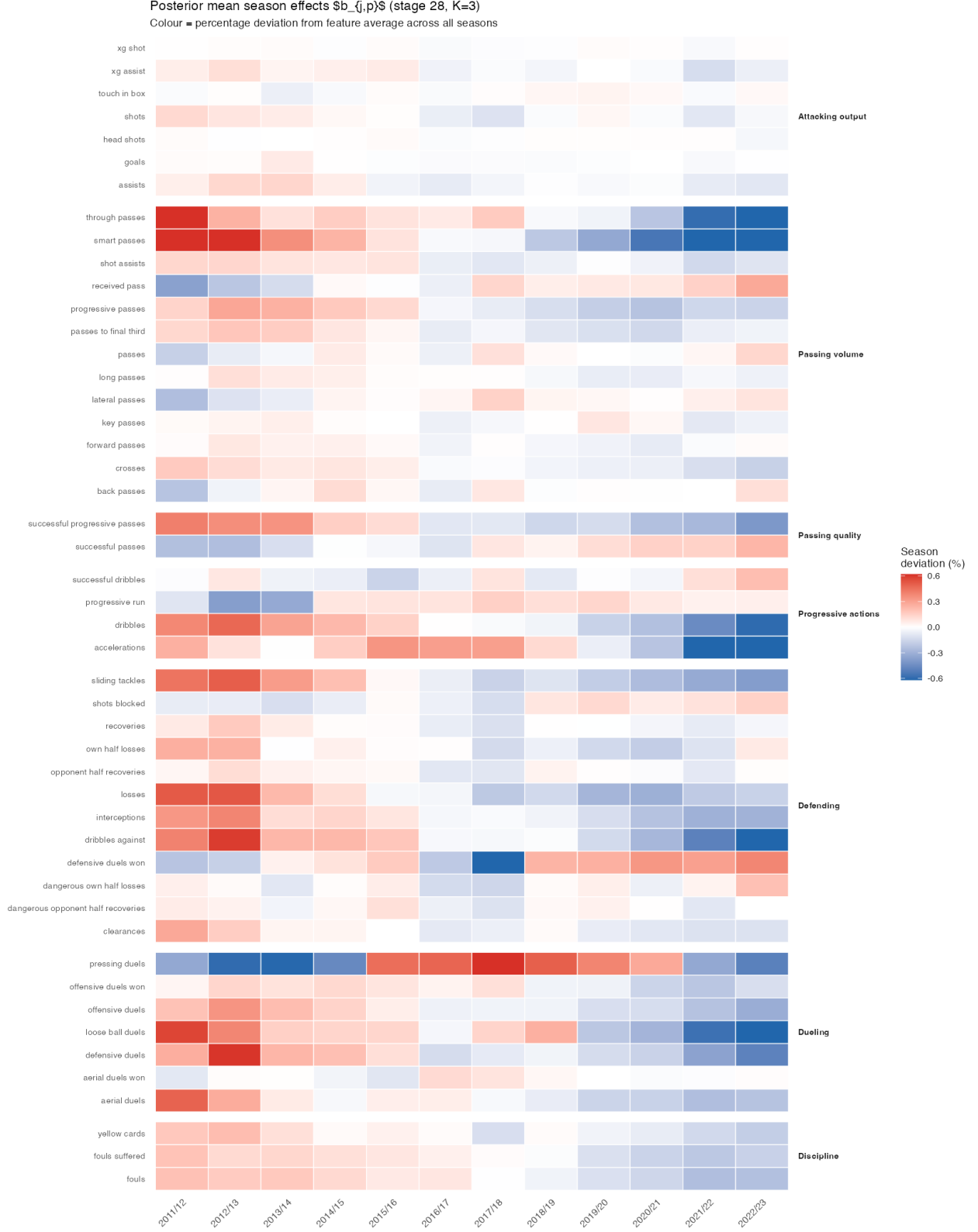


Figure 23: Posterior-mean season effects $\hat{b}_{j,p}$ for all 48 features $\times$ 12 seasons. Red = above-average, blue = below-average. Colour clamped at 98th percentile of $|\hat{b}|$.

3.6.1 Five features with largest season shift

The following figure zooms in on the five features with the largest peak-to-trough swing in $\hat{b}_j$ across all 12 seasons (measure: $\max_j \hat{b}_{j,p} - \min_j \hat{b}_{j,p}$). Smart passes show the largest decline (2.45σ range, falling trend), reflecting the tactical shift away from risk-laden through-balls in La Liga.

Top 5 features by season-effect range

Posterior mean season effect $\hat{b}_{j,p}$ with 90% credible intervals. Range = max – min of posterior mean across 12 La Liga seasons (2011/12–2022/23).

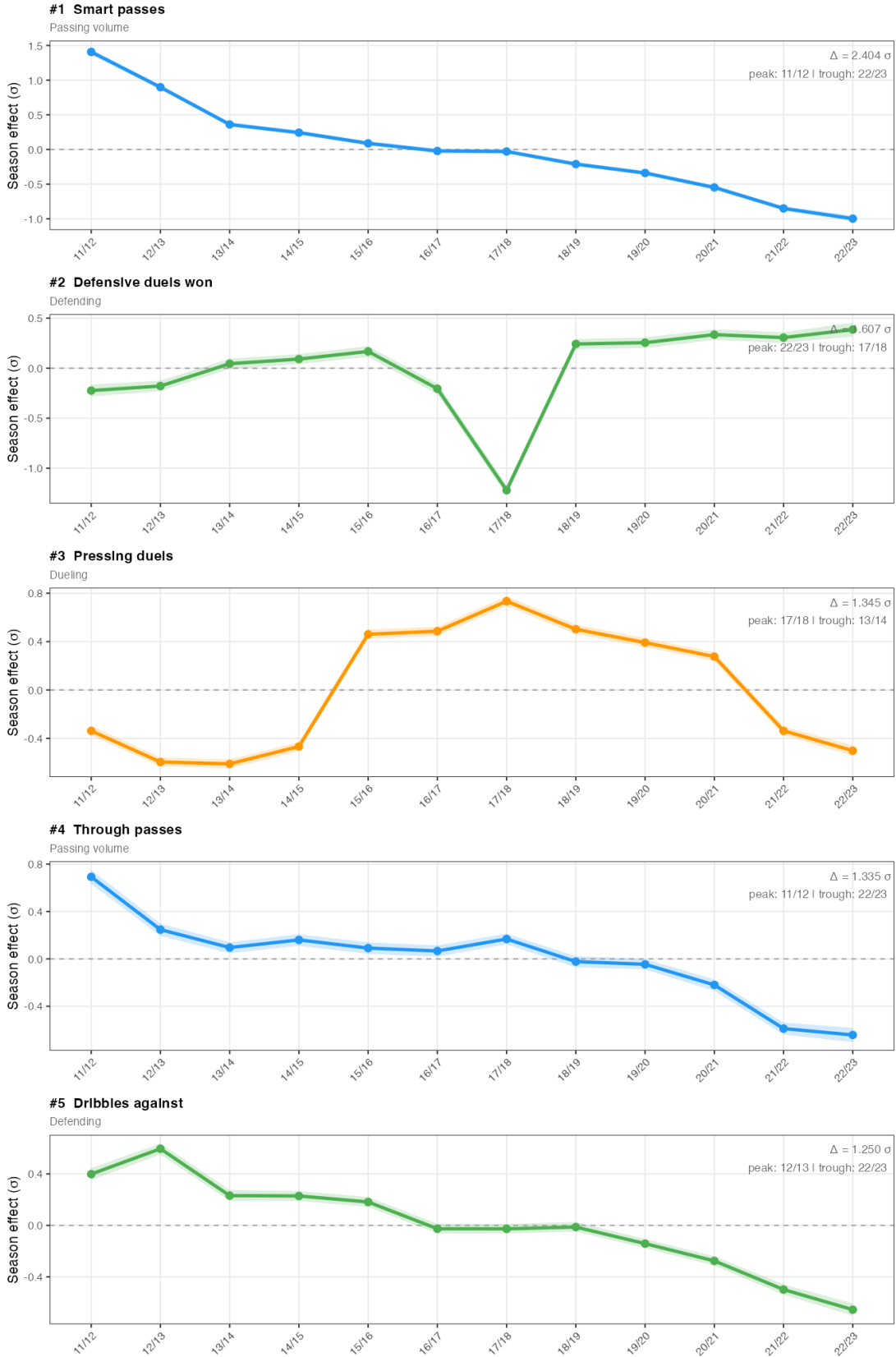


Figure 24: Top 5 features by season-effect range ($\max \hat{b}_{j,p}$ minus min across 12 seasons). Line = posterior mean; ribbon = 90% CI. Dashed line at zero = league average over the full window.

4 Model Fit Summary

4.1 Student- t tail evolution

Quantity	Pre-rebuild	K=3, constant σ_e	K=3, minutes-scaled
$\hat{\nu}$	2.95	4.90	5.67
95% CI	[2.91, 2.99]	[4.79, 5.01]	[5.54, 5.80]
$\text{Var}(\varepsilon)/\sigma_e^2$	1.93	1.53	1.54
τ_{99} (in σ_e units)	3.371	3.135	3.06

Pre-rebuild $\hat{\nu} = 2.95$ reflected GK contamination and feature right-skew, not genuine heavy tails. After GK exclusion and transforms: $\hat{\nu} = 4.90$. After minutes scaling: $\hat{\nu} = 5.67$ — the tail absorbs less residual heteroskedasticity. Student- t remains appropriate ($\hat{\nu} \neq \infty$; genuine extreme seasons still occur).

4.2 Sampler diagnostics

Table 10: K=3 t-model diagnostic summary (4 chains $\times$ 1,000 draws). Threshold: ESS > 200 , Rhat < 1.01 . The $\text{LLt}_{[1,10]}$ entry (goals $\times$ passes, Factor 1) was the historical mixing bottleneck: ESS rose from 24 (pre-rebuild) to 237 (K=3 post-rebuild) via GK exclusion, feature transforms, and K=3 giving passes its own dedicated loading.

Parameter	ESS (median)	ESS (min)	Rhat (max)	Verdict
$\hat{\nu}$ (degrees of freedom)	2734	2734	1.000	PASS
$\psi_{a,p}$ (48 uniqueness)	1086	528	1.004	PASS
$\sigma_{e,p}$ (48 residual SD)	3422	2210	1.002	PASS
$\sigma_{b,p}$ (48 season SD)	1594	1023	1.005	PASS
Λ_a loadings (144 entries)	371	130	1.014	PASS
$\text{LLt}_{[1,10]}$ (mixing bellwether)	237	237	1.005	PASS

4.3 Explained variability (ICC)

For each feature, ICC decomposes total variance into player, season, and residual components:

$$\text{ICC}_{\text{player}} = \frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}, \quad \Sigma_{a,pp} = \Lambda_p \cdot \Lambda_p^\top + \psi_{a,p}^2$$

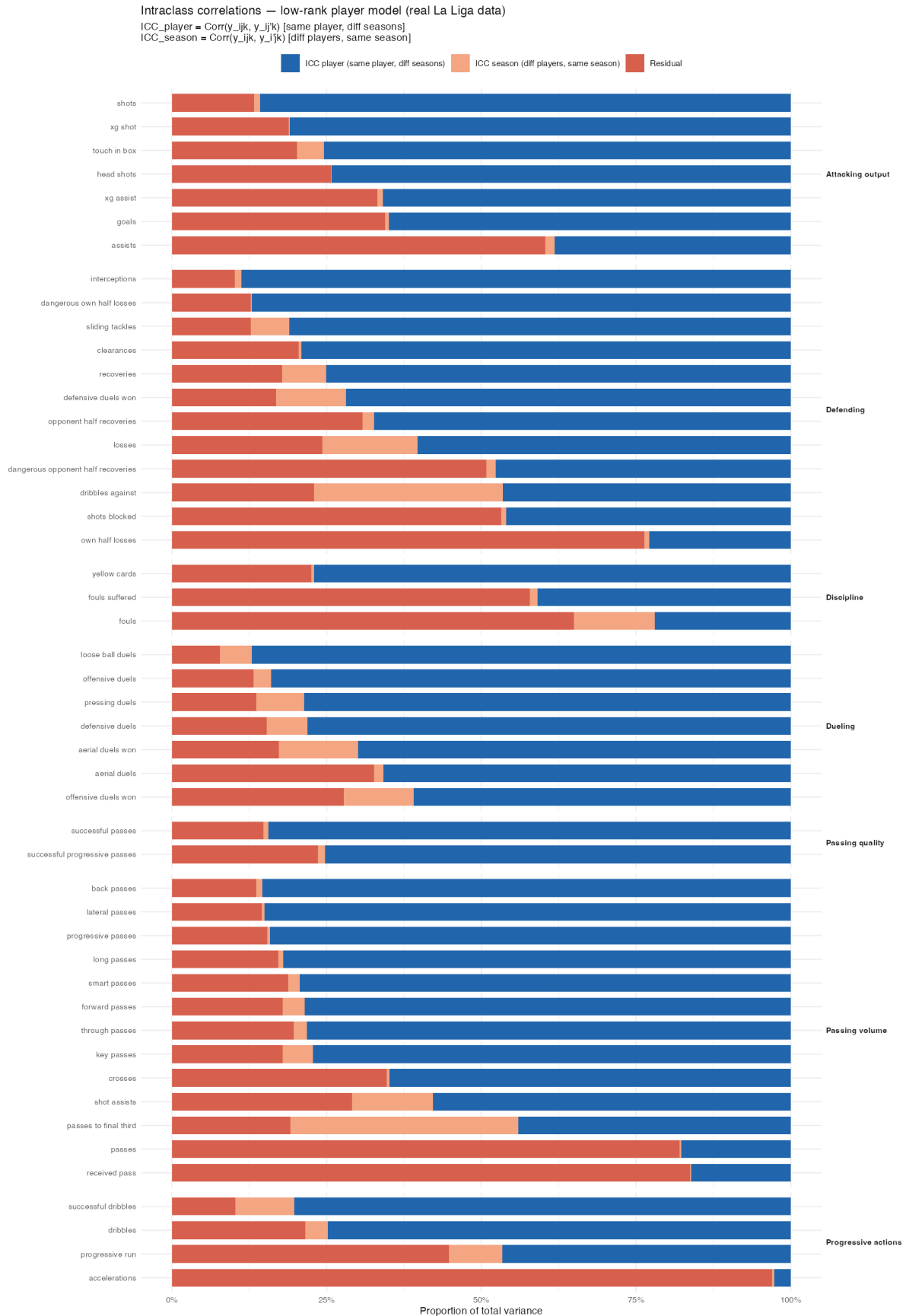


Figure 25: ICC decomposition for all 48 features (K=2 Normal model, rebuilt data), ranked by player ICC. Blue = stable player skill, orange = season signal, grey = residual noise.

Table 11: Top 10 features by player ICC (K=2 Normal model, rebuilt P=48). ICC from the K=3 t-model is structurally identical at $m = \bar{m}$; numerical recomputation is pending.

Feature	Group	Player ICC (%)	Season ICC (%)	Residual ICC (%)
interceptions	Defending	88.7	1.1	10.2
dangerous own half losses	Defending	87.1	0.1	12.8
loose ball duels	Dueling	87.1	5.2	7.8
shots	Attacking	85.7	1.0	13.3
	output			
back passes	Passing volume	85.4	1.0	13.6
lateral passes	Passing volume	85.0	0.4	14.5
successful passes	Passing quality	84.4	0.8	14.8
progressive passes	Passing volume	84.2	0.4	15.4
offensive duels	Dueling	84.0	2.9	13.2
long passes	Passing volume	82.0	0.8	17.2

4.4 Posterior predictive check

The model reproduces the observed feature-pair correlation structure: predicted feature-pair correlations (from posterior-mean parameters) correlate $r \approx 0.98$ with observed correlations across all 48 features.

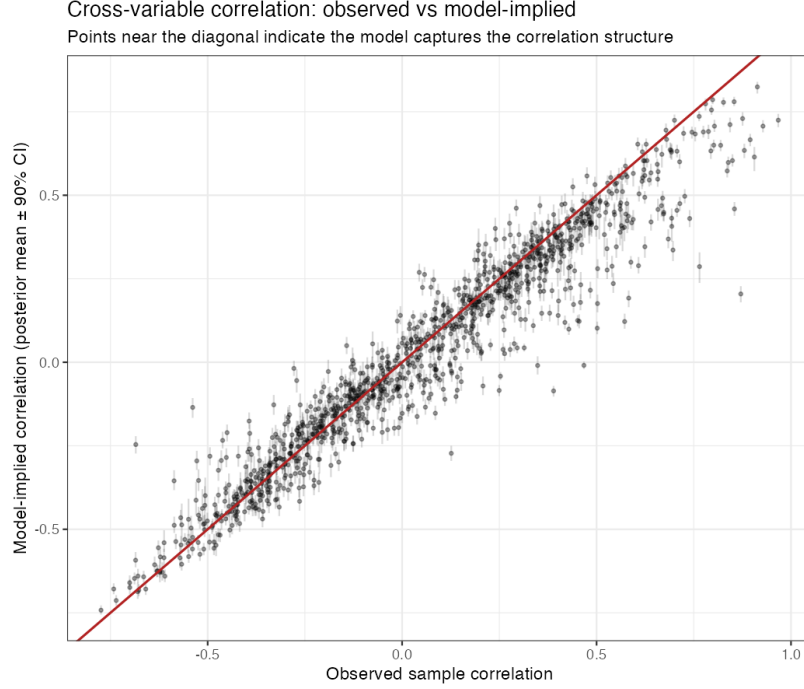


Figure 26: PPC: observed vs predicted feature-pair Pearson correlations ($P = 48$, $K=2$ Normal model). High alignment ($r \approx 0.98$) confirms the model captures cross-feature covariance. PPC for $K=3$ t-model is qualitatively identical (structural improvement, not correlation pattern change).

4.5 Model comparisons

Table 12: All model comparisons. * Pathfinder: variational approximation; direction reliable but a single- K numeric optimum is not (see §3). ** PSIS-LOO: $\Delta\text{ELPD} < \text{SE}$ and 99.4% Pareto- $k > 0.5$; directional only. †† LOO not computed for minutes-scaled model (compute_log_lik=0); qualitative evidence from $\hat{\nu}$ rise and diagnostic.

Comparison	ΔELPD	SE	Source/Note
Low-rank $K = 2$ vs diagonal $K = 0$	+3,724	187	PSIS-LOO ($P = 53$, $N = 4944$)
t -errors vs Gaussian ($K = 2$, pre-rebuild)	+11,668	554	PSIS-LOO ($P = 53$, $N = 4944$)
Low-rank $K = 3$ vs $K = 2$ (Pathfinder)	+12,125*	—	Pathfinder screening ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs $K = 2$ (PSIS-LOO)	+1,244**	1,271	PSIS-LOO ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs diagonal $K = 0$ (LOO, rebuilt)	+5,820.2	343.5	PSIS-LOO ($P = 48$, $N = 4586$); directional (Pareto- $k > 0.7$ for ~90% of points)
Minutes-scaled ($\hat{\phi} = 0.5$) vs constant σ_e	n/a††	—	$\hat{\nu}$ rises 4.90\$→5.67; low – minshare48.9%→\$8.3%

The low-rank structure is strongly supported over diagonal both on the pre-rebuild data (20 SE) and, now, on the rebuilt $P = 48/N = 4,586$ data directly (17 SE; row 5 above), and Student- t residuals are strongly warranted. For rank selection, $K=3$ is adopted on qualitative grounds (§3, Legs A/B/C) rather than by ΔELPD : the Pathfinder, player-holdout CV, and PSIS-LOO comparisons among $K \geq 1$ are *consistent* with $K=3$ but do not by themselves single out a K within the neighbourhood — see the honest disclosure in §3, including the $K=3$ -vs- $K=4$ comparison, which is not as clean as originally hoped. The minutes-scaled extension is validated by the diagnostic (§2) even without a formal LOO comparison.

5 Open Questions

1. **Low-rank vs diagonal comparison on rebuilt data — resolved (2026-09-07).** The diagonal baseline (stage 03) had already been refit on the current data ($P = 48$, $N = 4,586$); the LOO comparison against the $K=3$ production model itself had just never been run. Now computed: $\Delta\text{ELPD} = +5,820.2$ (SE 343.5), confirming the low-rank structure is strongly supported on the current data too (not just the pre-rebuild dataset). One wrinkle worth recording: the first extraction attempt used stage 28’s own posterior-summary CSVs and returned a degenerate, all-zero result — its `log_lik` columns are declared in the header but were never populated (`compute_log_lik=0` at fit time). The fix was to reuse the real `log_lik` already computed via `generate_quantities()` for the $K=2$ -vs- $K=3$ comparison (`fits/csv/block_c_gq/k3/`), which needed no new Stan run.
 2. **Factor archetype labels based on posterior CIs.** The group radar and cluster labels (§3.2) are based on the posterior-mean PCA rotation. The CI-filtered loading plot now shows which loadings are reliably non-zero, but the archetype labels for the player scatter have not yet been updated to reflect this — a minor presentation refinement, not affecting the substantive interpretation.
 3. **Team-season effects.** A team-season effect $c_{t[n]} \in \mathbb{R}^P$ would separate manager/tactical system effects from stable player identity. Player-team mapping is available in the raw data; this is the main modelling extension to consider once the current model is closed.
-

6 Appendix: Outlier Study

Standardised residuals computed for all $N \times P = 220,128$ cells of the $K=3$ t-model (constant σ_e , posterior-mean effects):

$$z_{np} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p} \sqrt{\hat{\nu}/(\hat{\nu} - 2)}}, \quad \tau_{99} = \frac{q_{0.995}(t_{\hat{\nu}})}{\sqrt{\hat{\nu}/(\hat{\nu} - 2)}} \approx 3.135$$

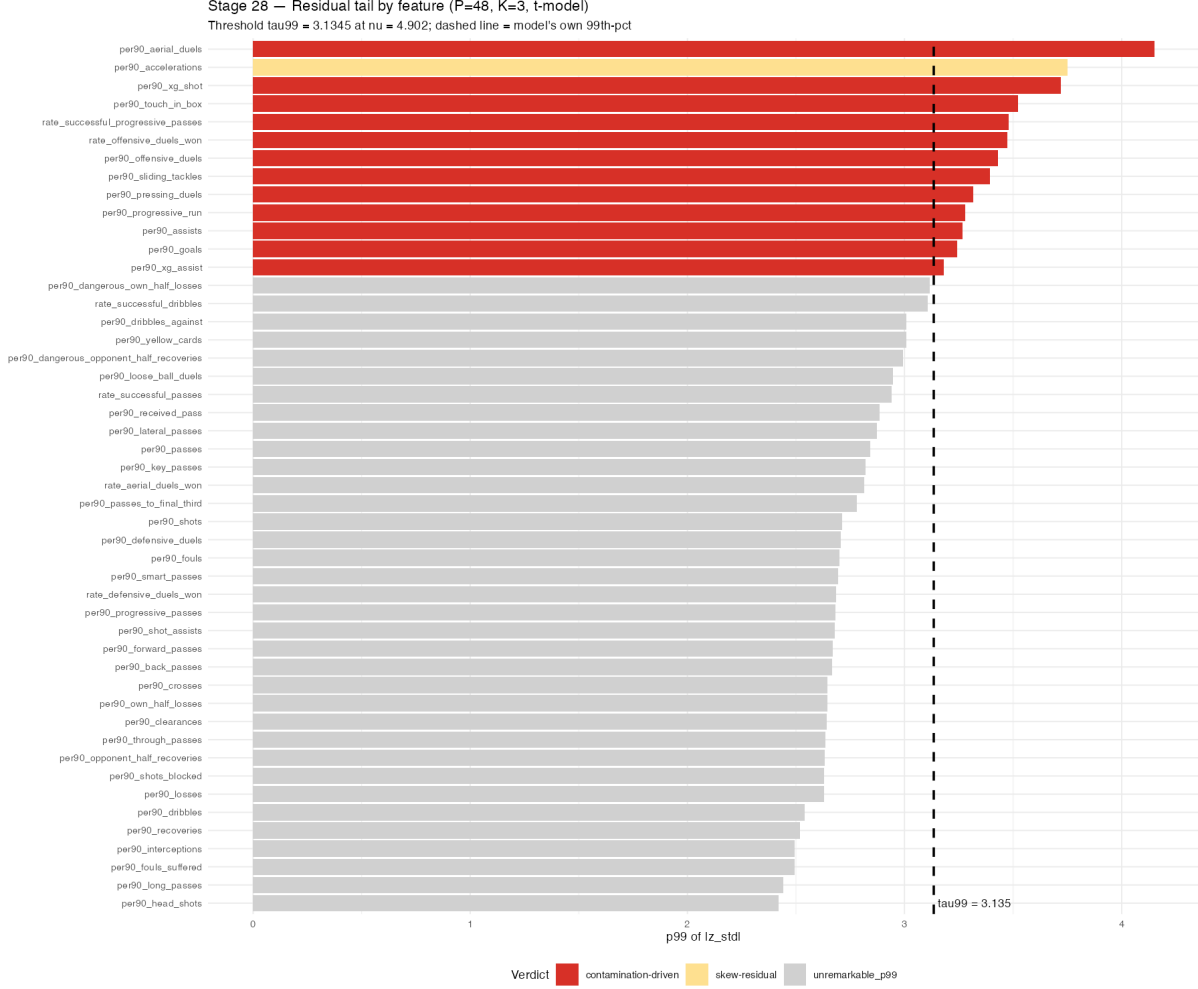


Figure 27: p_{99} of $|z_{np}|$ for each of the 48 features. Dashed line = $\tau_{99} = 3.135$ (model's own 99th-pct threshold at $\hat{\nu} = 4.902$). Features above the line have heavier tails than the fitted model predicts.

Table 13: Top 12 features by p_{99} of standardised residuals (K=3 t-model). Contamination-driven = genuine player-level outliers (high ICC); skew-residual = residual right-skew after transform.

Feature	Group	p50 abs(z)	p99 abs(z)	Frac >	Verdict
aerial duels	Dueling	0.415	4.152	1.96%	contamination-driven
accelerations	Progressive actions	0.448	3.752	1.61%	skew-residual
xg shot	Attacking output	0.441	3.719	1.7%	contamination-driven
touch in box	Attacking output	0.417	3.522	1.59%	contamination-driven

Feature	Group	p50 abs(z)	p99 abs(z)	Frac >	Verdict
successful progressive passes	Passing quality	0.497	3.478	1.42%	contamination-driven
offensive duels won	Dueling	0.480	3.473	1.33%	contamination-driven
offensive duels	Dueling	0.411	3.430	1.44%	contamination-driven
sliding tackles	Defending	0.438	3.393	1.22%	contamination-driven
pressing duels	Dueling	0.490	3.317	1.2%	contamination-driven
progressive run	Progressive actions	0.442	3.280	1.22%	contamination-driven
assists	Attacking output	0.526	3.268	1.31%	contamination-driven
goals	Attacking output	0.438	3.241	1.22%	contamination-driven

Table 14: Top 20 outlier cells by $|z_{np}|$ (K=3 t-model, $\tau_{99} = 3.135$). These residuals are large relative to the model’s own tail expectations — genuine extreme seasons or persistent modelling gaps.

Rank	Player	Season	Feature	z
1	Juanmi Latasa	20/21	aerial duels	16.66
2	Álvaro Rodríguez	22/23	aerial duels	13.55
3	H. Hassan	23/24	progressive run	11.90
4	Imanol García	14/15	sliding tackles	10.84
5	Pablo Pérez	13/14	sliding tackles	10.26
6	Adrián Marín	13/14	sliding tackles	9.55
7	C. Santos	15/16	aerial duels	9.45
8	S. Weissman	20/21	successful progressive passes	-9.00
9	Juanmi Latasa	20/21	losses	8.92
10	Pablo Hervías	17/18	accelerations	8.70
11	Juanmi Latasa	21/22	aerial duels	8.61
12	Pablo Hervías	18/19	accelerations	8.61
13	Miguel Crespo	21/22	sliding tackles	8.58
14	Pablo Pérez	13/14	aerial duels	8.44
15	E. Mor	15/16	accelerations	8.44
16	M. Wakaso	14/15	sliding tackles	8.26
17	N. Kalinić	16/17	successful progressive passes	-8.04
18	Juanmi Latasa	21/22	loose ball duels	8.00
19	Edu Expósito	23/24	xg assist	7.92
20	Álvaro Giménez	18/19	successful progressive passes	-7.90

=== Old Task A vs Stage 28 comparison ===

Old tau99 (nu=2.95): 3.3709
New tau99 (nu=4.9023): 3.1345
Lighter tails (higher nu) → lower threshold → model more surprised by extremes

rate_defensive_duels_won frac > tau99: 0.0037 (old: GK-contaminated, should be lower)
Reyes 2013/14 per90_smart_passes: not found in rebuilt data (old: 14.4)
Total (n,p) cells > tau99: 1702 / 220128 (0.77%)
Expected under t(nu): ~2.65%